Impacts of Secondary School in Kenya after Two Decades

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Abstract

We study the long-run effects of access to secondary school in Kenya, where admission depended on a cutoff score on a national exam at the end of primary school. Using a regression discontinuity design and a panel that follows students for nearly two decades after the exam, we estimate effects on fertility, cognition, and economic outcomes. Scoring above the cutoff raises secondary school completion by 16 percentage points, with larger increases for women. We find large and persistent declines in realized fertility among men, of about one child on average, but, perhaps surprisingly, no detectable fertility effects among women. These fertility changes are not accompanied by changes in the probability of marriage. We also document durable improvements in adult cognitive performance, with gains of roughly 0.2–0.3 standard deviations. These benefits appear to extend across generations: the children of female respondents who scored above the cutoff perform better on tests of academic skills, while the children of male respondents do not. In contrast, we find limited evidence of long-run gains in earnings or household consumption.

JEL codes: I21, J31, O12, O15.

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1 Introduction

As part of the Sustainable Development Goals, the focus of global education policy has shifted toward secondary schooling (Duflo et al., 2026). About half of the countries in Sub-Saharan Africa now offer fee-free public secondary school (Garlick, 2019; Ferreira and Sandholtz, 2024). In Kenya, our study setting, the government abolished tuition fees in public secondary schools in 2008, following the 2007 presidential election (Ohba, 2011; Brudevold-Newman, 2021). Yet credible evidence on the long-run effects of secondary schooling in low- and middle-income countries (LMICs) remains scarce, because expansions of secondary schooling typically reach entire cohorts at once and few studies follow students far into adulthood.

This paper studies the long-run effects of access to secondary school in Kenya. Under Kenya's former school system, admission to government secondary schools depended on scores on the Kenya Certificate of Primary Education (KCPE).[1] The probability of admission rose sharply at a particular score, allowing us to compare students who scored just above and just below the cutoff. Ozier (2018) used this discontinuity to study the same cohort in early adulthood and found gains in vocabulary and reasoning, a reduction in low-skill self-employment among men, and a reduction in teen pregnancy among women. We extend this analysis by nearly two decades, following the same individuals into their late 30s.

We link administrative KCPE records to the Kenya Life Panel Survey (KLPS), which has followed approximately 7,500 individuals from rural western Kenya since they were enrolled in primary school in 1998.[2] Improved matching to the exam records expands the analysis sample by about 1,000 individuals relative to Ozier (2018), to 3,162 respondents. The most recent round administered an adaptation of the Harmonized Cognitive Assessment Protocol (HCAP), a cognitive battery covering memory, language, executive function, orientation, and

[1] Kenya has since moved from the 8–4–4 system to a competency-based curriculum (Akala, 2021). The last KCPE was administered in 2023 (Kalekye, 2023), and placement into senior school is now based on the Kenya Junior School Education Assessment, which pupils first took in 2025 at the end of junior school (Ayega, 2025).

[2] The effective tracking rate was 84 percent in the 2017–2019 survey round, and rates were similar in the 2007–2009 and 2011–2014 rounds (Hamory et al., 2021).

visuospatial ability (Gross et al., 2025). The survey rounds also record fertility, marriage, earnings, and household consumption. To measure outcomes for the next generation, the most recent round assessed up to two of each respondent's children aged 8 to 14 in language, math, executive function, and reasoning.

Crossing the KCPE cutoff substantially increases educational attainment. Scoring above the cutoff raises the probability of completing secondary school by 16 percentage points, from 30 percent just below the cutoff, and adds about 0.7 years of schooling. This is close to the 15 percentage point increase that Ozier (2018) estimated in the original sample. The increase in completion is larger for women (22 percentage points) than for men (11 percentage points), although the estimate for men is imprecise. Most regression discontinuity studies of school admission compare students admitted to more and less selective schools, nearly all of whom attend school, and therefore estimate the effect of school quality (Jackson, 2010; Pop-Eleches and Urquiola, 2013; Abdulkadiroğlu et al., 2014; Lucas and Mbiti, 2014). In our setting, by contrast, 43 percent of students below the cutoff never attend secondary school, so the comparison captures the effect of access to secondary school itself. Because the exam was graded centrally, students could not precisely manipulate their scores, and students on either side of the cutoff have similar background characteristics and are similarly likely to be found in later survey rounds, which supports a causal interpretation of these comparisons.

We find large and lasting gains in cognition. At an average age of 37, respondents just above the cutoff score 0.23 standard deviations higher than respondents just below it on a general cognition factor that summarizes performance across all cognitive domains. Among the individual domains, the gain is largest for memory (0.28 standard deviations), while estimates for language, executive function, orientation, and visuospatial ability are positive but not statistically significant. These effects are large given that the cutoff changes secondary completion for only about one in six students. Scaled by the first stage, they imply that each additional year of schooling raises general cognition by about 0.3 standard deviations. Together with the gains in vocabulary and reasoning that Ozier (2018) found in

respondents' early 20s, our results show that the skills students gained in secondary school persisted for nearly two decades, into midlife. Despite widespread concern that expanding access to school in LMICs produces little learning (Pritchett, 2013; World Bank, 2018), these results indicate that secondary school in this setting built lasting human capital.

We do not find effects on earnings or household consumption. In data pooled over four survey rounds from respondents' early 20s to their late 30s, the estimated effect on annual non-farm earnings is −105 dollars relative to a below-cutoff mean of 2,187 dollars (2024 USD PPP), and the effect on per capita household consumption is also small. These estimates are imprecise, however, and the 95 percent confidence interval for earnings, which runs from 37 percent below to 22 percent above the mean, includes moderate gains. One reason may be that formal wage jobs, where the returns to education in Africa are concentrated, are scarce in this setting, particularly for women (Rankin et al., 2010; Byambaa et al., 2025).

Most research on schooling and family formation focuses on women, but in our setting the long-run decline in fertility occurs among men. By their late 30s, men just above the cutoff have about one fewer child than men just below it, a 31 percent decline, even though the increase in schooling at the cutoff is larger for women. The gap emerges in their mid-20s and widens over time. Women's fertility, in contrast, is unchanged by their late 30s, and we can rule out declines larger than 0.56 children, which suggests that the reduction in teen pregnancy found by Ozier (2018) reflected delayed childbearing rather than fewer births. Desired family size falls by 0.3 to 0.5 children for both men and women. The probability of marriage and the number of marriages do not change at the cutoff, but men above the cutoff marry women with about 1.2 fewer years of schooling, while the spouses of women above and below the cutoff are similar.

We also find suggestive evidence that the gains in human capital pass to the next generation through mothers. Children aged 8 to 14 whose mothers scored just above the cutoff perform 0.36 standard deviations better on an index of academic skills, with the largest gain in language (0.41 standard deviations), while the children of male respondents show

no gains. These estimates are significant at the 10 percent level, and the set of children we observe depends partly on fertility, which itself responds to the cutoff, so we view them as suggestive. Causal evidence on whether parents' schooling affects their children's skills is limited, particularly in LMICs, where most estimates come from expansions of primary schooling. Our estimates suggest that the returns to secondary schooling for girls extend to the next generation, so evaluations based only on students' own outcomes would understate them, with direct implications for the case for public subsidies of secondary school (Duflo et al., 2026).

Our paper contributes to three literatures. First, it adds to the limited causal evidence on the long-run effects of secondary schooling in LMICs. The closest study is a randomized evaluation of secondary school scholarships in Ghana, which followed students into their early 30s (Duflo et al., 2025, 2026). The two designs study different students. The Ghana scholarships went to students who had already been admitted to secondary school but could not pay the fees, so that study measures the effect of removing a financial barrier for academically qualified students. The KCPE cutoff instead determines admission itself, so our estimates capture the effect of secondary school for students of average academic performance whose admission depended on a few points on the exam. Despite these differences, the findings are similar in several respects. Both studies find gains in measured skills, little evidence of earnings gains among men, and improvements among the children of women but not men. In both settings, schooling appears to delay childbearing among women without reducing the number of children they have by the latest follow-up. Our study follows respondents further into adulthood, allowing us to examine cognition at an average age of 37 and the academic skills of older children, aged 8 to 14. The longer follow-up also reveals a substantial decline in men's fertility, which is difficult to assess in Ghana, where men in the control group had fewer than one child on average by about age 30. Long-run follow-ups of cash transfer experiments in Malawi similarly find lasting effects on schooling, marriage, and fertility but not on earnings (Baird et al., 2011, 2019).

The longer follow-up also changes what we learn about fertility. Most evidence on education and family formation in LMICs concerns marriage and births in adolescence or early adulthood (Breierova and Duflo, 2004; Osili and Long, 2008; Chicoine, 2012; Duflo et al., 2015; Kırdar et al., 2018; Keats, 2018; Brudevold-Newman, 2021; Kazibwe and Li, 2025). In Turkey, for example, the effects of a compulsory schooling reform on teen marriage and births disappear by the late teens (Kırdar et al., 2018). Our results for women show that early effects of this kind can reflect delayed childbearing rather than fewer births over the life course, consistent with evidence from compulsory schooling reforms in Europe (Monstad et al., 2008; Fort et al., 2016), although a stipend program in Bangladesh did reduce fertility (Hahn et al., 2018). Our results for men show that effects on fertility can emerge later in life, at ages that studies of early fertility, which mostly examine women, do not observe.

Second, we find that the cognitive gains from secondary schooling in Kenya persist into midlife. Causal evidence on schooling and later-life cognition comes mostly from compulsory schooling reforms in Europe and the United States, which find that schooling improves memory decades later (Glymour et al., 2008; Banks and Mazzonna, 2012; Schneeweis et al., 2014), and a meta-analysis finds that each year of schooling raises IQ by 1 to 5 points (Ritchie and Tucker-Drob, 2018). Our estimate of about 0.3 standard deviations per year falls within this range. Evidence from LMICs is more limited and mixed, and it mostly measures skills soon after school or at the primary margin (Filmer and Schady, 2014; Ozier, 2018; Duflo et al., 2025; Amin et al., 2025). Effects on midlife cognition also matter for the region's aging population, because low educational attainment is a leading modifiable risk factor for dementia (Livingston et al., 2020) and the number of older adults in Sub-Saharan Africa is projected to triple over the next three decades (Duhon et al., 2023).

Third, our results contribute to the small literature on the causal intergenerational effects of parental education. Studies of schooling reforms and college access in Europe and the United States find positive but mixed effects of parents' schooling on children (Oreopoulos et al., 2006; Currie and Moretti, 2003; Carneiro et al., 2013), and some find effects only of

mothers' schooling (Black et al., 2005; Lundborg et al., 2014). Evidence from LMICs comes mostly from expansions of primary schooling and other education reforms (Breierova and Duflo, 2004; Akresh et al., 2023; Güneş, 2015; Grépin and Bharadwaj, 2015; Agüero and Ramachandran, 2020; Wantchekon et al., 2015; Andrabi et al., 2012) and from childhood health interventions in the KLPS cohort (Walker et al., 2023; Duhon et al., 2024). In Sub-Saharan Africa, the closest evidence on secondary schooling comes from the Ghana experiment, which measured children's cognitive development up to age 7 (Duflo et al., 2026). We extend this evidence to children's academic skills at ages 8 to 14. Our results also suggest that the intergenerational returns to secondary schooling depend on whether it raises the schooling of mothers. Gains appear only among the children of women, while men above the cutoff marry less-educated women, so the mothers of their children have, if anything, less schooling.

The remainder of the paper describes the Kenyan secondary-school system and the KLPS data, presents the regression-discontinuity design, and then examines effects on education, cognition, fertility and marriage, economic outcomes, and the next generation.

2 Background: Secondary Schooling in Kenya

The cohorts in our study were educated under Kenya's 8–4–4 system, which consisted of eight years of primary school (Standards 1 to 8) followed by four years of secondary school (Forms 1 to 4). At the end of primary school, pupils took the Kenya Certificate of Primary Education (KCPE), a national examination used to determine admission to government secondary schools (Ozier, 2018). The exam was graded centrally, and a score of 250 out of 500 was widely regarded as a passing grade (Akolo, 2008; Ozier, 2018).[3]

Admission thresholds varied across schools and by gender. More selective secondary schools required higher scores (Lucas and Mbiti, 2014; Ozier, 2018). In a 2009 survey of

[3]For pupils who took the KCPE in 1999–2000, the examination was scored out of 700 points. We rescale these scores to the 500-point scale used in subsequent years by multiplying by 5/7.

secondary schools in the study area, no school reported an admission cutoff below 250 for boys, but 7 of 18 schools reported a cutoff below 250 for girls (Ozier, 2018). These thresholds were not strictly binding. Students who scored above 250 were not guaranteed admission, since the number of qualifying candidates could exceed the available places (Aduda, 2008; Akolo, 2008). Those who were not admitted could repeat Standard 8 and retake the KCPE, enroll in a private secondary school, or pursue vocational training (Ozier, 2018).

Even students admitted to government secondary schools could not always afford to attend. Before 2008, tuition and other fees at even the least selective government secondary schools amounted to hundreds of dollars per year (Ozier, 2018). In 2008, following the 2007 presidential election, the government abolished tuition fees in public secondary schools, although families continued to face other schooling costs (Ohba, 2011). Our study cohorts took the KCPE between 1999 and 2006, before the fee reform, when access to secondary school was constrained by both admission requirements and costs.

Although there was no single admission cutoff for all students, the probability of entering government secondary school rose sharply at particular scores. Following Ozier (2018), we identify these cutoffs separately by gender, yielding 250 points for boys and 234 points for girls (Section 4). Because more selective schools set higher cutoffs, these thresholds mark the margin of admission to the least selective government schools. At these thresholds, the relevant distinction is therefore primarily between attending secondary school and not attending, rather than between schools of different quality. Because admission was neither guaranteed above the cutoff nor impossible below it, crossing the threshold increased the probability of secondary schooling without determining whether a student attended. We therefore report reduced-form effects of scoring above the cutoff, and because students below it could retake the exam, we use each student's first score.

3 Data

3.1 Kenya Life Panel Survey

The primary data source for this study is the Kenya Life Panel Survey (KLPS), a longitudinal survey that has followed approximately 7,500 individuals who attended 75 primary schools in Busia District, in rural western Kenya, in 1998 (Hsu et al., 2026). These schools participated in the Primary School Deworming Project, a randomized evaluation of school-based deworming (Miguel and Kremer, 2004). The KLPS has completed four survey rounds (2003–2005, 2007–2009, 2011–2014, and 2017–2022), with effective tracking rates of 80 to 85 percent in each round. A fifth round (KLPS-5) began in 2023 and is ongoing (Hsu et al., 2026).[4]

Across survey rounds, the KLPS has collected detailed information on educational attainment, employment, earnings, household consumption, marriage, and fertility. Because the survey follows respondents from childhood into midlife, it can trace how the effects of secondary schooling evolve over adulthood, including whether early changes in employment and fertility persist as respondents grow older.

An important addition in KLPS-5 is a detailed assessment of adult cognitive function through the Cognitive Plus (C+) module. The module administered an adaptation of the Harmonized Cognitive Assessment Protocol (HCAP), a battery of neuropsychological tests developed for the U.S. Health and Retirement Study and used in comparable studies of aging across countries. The KLPS version was adapted to the Kenyan context, and Gross et al. (2025) show that it measures cognition reliably in this population. The assessment covers five domains: orientation, memory, executive function, language, and visuospatial ability.[5] We

[4]KLPS-5 data are collected through several survey modules, each fielded in two representative waves. This paper uses the Cognitive Plus (C+) module, the Integrated (I+) module, and the Kids Assessments and Primary Caregiver (PC) modules, all of which are complete. KLPS-5 earnings come from the C+ module. The Earnings Plus (E+) module, which collects more detailed earnings and consumption data, is still ongoing.

[5]Orientation is assessed with questions on time and place. Memory tests include the CERAD 10-Word List Learning task, the East Boston Memory Test, and a Logical Memory test. Executive function tests include Raven's Progressive Matrices, the Token Test, Go/No-Go, and digit span. Language tests include

use the general cognition and domain-specific factor scores constructed by Gross et al. (2025) using confirmatory factor analysis, standardized to have mean zero and standard deviation one in the full KLPS-5 cognition sample. These assessments provide a more comprehensive measure of cognitive function than the vocabulary and reasoning tests used by Ozier (2018) in early adulthood, allowing us to examine whether cognitive gains are still evident nearly two decades after respondents took the KCPE. In our analysis sample, 2,451 respondents completed the cognition module.

KLPS-5 also assessed respondents' children, providing measures of outcomes for the next generation. Up to two biological children of each respondent, one younger and one older, were selected at random in the previous survey round (Hsu et al., 2026).[6] At the KLPS-5 assessment, they were aged 8 to 14. The assessments include tests of vocabulary, verbal fluency, reading, mathematics, working memory, executive function, and nonverbal reasoning.[7] Test scores are standardized separately by age group and survey wave and combined into four indices measuring language, math and spatial ability, executive function, and fluid intelligence. The main outcome is an academic index combining language and math. Because these children are of school age, we can measure their performance in reading and mathematics alongside broader cognitive skills. The analysis sample includes 2,045 assessed children.

3.2 Administrative Data

The KLPS does not record official exam scores, and self-reported KCPE scores are often misreported (Ozier, 2018). Respondents are therefore linked to an archive of 17,384 official KCPE records that Ozier (2018) collected from District Education Offices and, where those

animal fluency and object naming, and visuospatial tests include CERAD Constructional Praxis, copying interlocking pentagons, and clock drawing (Gross et al., 2025).

[6]Children were selected when KLPS-4 began, in 2018 for the first survey wave and in 2020 for the second. For each respondent, at most one child was selected from among biological children aged 2.5 to 5.5 years at that time and at most one from among those aged 5.5 to 8.5 years (Hsu et al., 2026).

[7]The assessments include the Peabody Picture Vocabulary Test (PPVT, Dunn and Dunn, 2007), an animal naming test, the Uwezo Kiswahili, English, and math tests (Africa, 2021), forward and backward digit span, and Raven's Progressive Matrices.

offices lacked records, directly from primary schools. The archive covers the years in which cohort members would have taken the KCPE (1999–2006) and records each candidate's name, sex, exam year, primary school, and total score. Because some paper records were lost, the archive covers 88 percent of the records sought (Ozier, 2018).

Respondents are matched to these records using a new procedure that compares every name a respondent has reported across KLPS rounds with the candidate's name, requires agreement on primary school, and checks that the sex and exam year are consistent with the respondent's survey data (Appendix B). Only matches identified automatically by this procedure are accepted. When a respondent matches records from more than one exam year, the analysis uses the earliest score consistent with the respondent's schooling history, which for students who retook the exam is their first attempt. The procedure yields 3,259 matches. After dropping 58 matches with implausible exam years and 39 records with missing scores, the analysis sample includes 3,162 respondents (1,397 women and 1,765 men), about 1,000 more than in Ozier (2018).

Appendix Table A1 compares the analysis sample with the full KLPS cohort. Because it includes only respondents who completed primary school, sat the KCPE, and could be matched to an official exam record, the analysis sample is more educated than the full cohort (72 percent attended secondary school, compared with 53 percent) and includes a smaller share of women.

Appendix Figure A2 plots the distribution of KCPE scores by gender in the full archive and in the matched sample. The matched sample closely tracks the archive, and both distributions are smooth. The cutoffs described in Section 2 fall near the middle of these distributions, so many students scored close to them. The female cutoff of 234 is at about the 45th percentile of female scores in the archive (52nd in the matched sample), and the male cutoff of 250 is at about the 41st percentile of male scores (47th in the matched sample).

4 Empirical Strategy

4.1 Regression Discontinuity Design

We use a regression discontinuity (RD) design to estimate the long-run effects of scoring above the KCPE cutoff for admission to government secondary school. Students who completed secondary school had higher exam scores and came from more advantaged backgrounds than those who did not, so differences in their later outcomes cannot necessarily be attributed to schooling. Near the admission cutoff, however, small differences in exam scores generated large differences in access to secondary school. We exploit this discontinuity by comparing students who scored just above and just below the cutoff.

Following Ozier (2018), we identify the admission cutoffs separately by gender using the structural break approach of Card et al. (2008). For each gender, we regress secondary school completion on the KCPE score, allowing for a discontinuity at a candidate cutoff and for different slopes on either side, and select the candidate cutoff that maximizes the R^2. This procedure yields 234 points for women, the same cutoff as in Ozier (2018), and 251 points for men.[8] Repeating the search separately for groups of exam years produces less precise estimates and does not identify clear alternative cutoffs (Appendix Figure A3), so we use a single cutoff for each gender throughout.

Let K_i denote respondent i's first KCPE score minus the cutoff for the respondent's gender, and let $D_i = \mathbb{1}\{K_i \geq 0\}$ indicate scoring at or above the cutoff. We estimate local linear regressions of the form

$$Y_{it} = \alpha + \tau D_i + \beta_1 K_i + \beta_2 (K_i \times D_i) + \gamma F_i + \lambda_t + \varepsilon_{it}, \tag{1}$$

where Y_{it} is an outcome for respondent i in survey round t, F_i is an indicator for women, and λ_t denotes survey-round fixed effects, included only for outcomes observed in several rounds.

[8] We use 250 points for men, the cutoff widely reported in national media and the most common cutoff reported by secondary school administrators in 2009 (Ozier, 2018). Appendix Figure A1 presents the structural break estimates.

Because the pooled sample combines two gender-specific cutoffs, F_i plays the role of the cutoff fixed effects used in RD designs that pool several cutoffs. It is omitted from the separate estimates for women and men. The coefficient τ measures the discontinuity in the outcome at the admission cutoff. We use a triangular kernel and the mean squared error optimal bandwidth for each outcome, selected by `rdrobust` (Calonico et al., 2014, 2019), and report conventional local linear point estimates with robust bias-corrected standard errors and p-values.[9] Estimates using bandwidths from half to twice the MSE-optimal bandwidth have the same sign and similar magnitudes, although several are smaller at the widest bandwidths (Appendix Figure A4). For outcomes observed in several rounds, we pool the observations and cluster standard errors by respondent. We report estimates for the pooled sample, with scores centered at the gender-specific cutoffs, and separately for women and men.

The identifying assumption is that, in the absence of the admission rule, potential outcomes would vary smoothly with KCPE scores at the cutoff. Under this assumption, τ identifies the causal effect of scoring above the admission threshold for students with scores close to it.

Because scoring above the cutoff increased the probability of secondary schooling without guaranteeing admission or completion, the design is a fuzzy RD. Our main estimates are reduced-form effects of crossing the cutoff rather than instrumental variables (IV) estimates of the effect of completing secondary school. An IV interpretation would require that scoring above the cutoff affect later outcomes only through secondary school completion, but admission could also change students' expectations, peer groups, or parental investments, even among students who do not complete secondary school. The first stage is also relatively small, raising completion by 16 percentage points in the pooled sample and by an imprecisely estimated 11 percentage points among men, so scaling the reduced-form estimates by the first stage would produce imprecise IV estimates, particularly for men. For the cognitive out-

[9] Although KCPE scores were originally recorded as integers, rescaling scores from 1999–2000 and centering at gender-specific cutoffs yields more than 180 distinct score values within the bandwidths used in the pooled analysis and more than 100 within each gender.

comes, we also report fuzzy RD estimates that instrument for secondary school completion with the cutoff, following Ozier (2018) (Panel C of Table 2).

Our estimates apply to students near the admission thresholds, which fall close to the middle of the KCPE score distribution (Section 3). They capture the effects of crossing the admission cutoff for cohorts who applied to secondary school before the 2008 tuition reform, rather than the effects of secondary schooling for all students.

4.2 Validity of the RD Design

We assess the validity of the RD design by examining the distribution of KCPE scores, balance in predetermined characteristics, matching to administrative records, and survey participation.

The KCPE was graded centrally, which makes it unlikely that students could precisely manipulate their scores around the admission thresholds. Consistent with this, density tests find no evidence of a discontinuity in the distribution of KCPE scores at the cutoff, either in the pooled sample or separately by gender (Appendix Figure A5). Predetermined characteristics, including year of birth, grade in 1998, parents' education, assignment to the deworming program, and baseline school characteristics, are also balanced at the cutoff for both women and men (Appendix Table A2).

A further concern is that the analysis includes only respondents whose KLPS records could be linked to official KCPE scores. If the probability of a successful match changes at the cutoff, the matched sample could differ systematically on either side of it. We examine this possibility using the full KCPE archive, testing whether the probability that an exam record is matched to a KLPS respondent changes at the cutoff (Appendix Table A4). Match rates are smooth at the cutoff for men and in the pooled sample, but fall by 5 to 6 percentage points for women.

One possible explanation is repeat exam-taking. Students who narrowly missed the cutoff could repeat Standard 8 and retake the KCPE, and those who did often scored above the

threshold on their second attempt. Because the matching procedure retains each respondent's earliest exam record, these later attempts would appear in the archive as unmatched records just above the cutoff. Consistent with this explanation, apparent repeat records lie mostly above the cutoff and are rarely matched (2.5 percent, compared with 19 percent of all records). The estimated discontinuity in match rates for women also becomes smaller and is no longer statistically significant when the scores closest to the cutoff are excluded. The distribution of scores in the matched sample is also smooth at the cutoff for both women and men, as are the predetermined characteristics of matched women. Appendix C reports these checks in more detail. Because of the lower match rate for women, the share of women in the analysis sample also falls at the pooled cutoff, by 6.5 percentage points ($p = 0.08$). We control for gender in the pooled estimates and report separate estimates for women and men.[10]

Finally, survey participation is similar for respondents just above and just below the cutoff in most rounds (Appendix Table A3). There are two exceptions. Participation in KLPS-2 is about 7 percentage points lower above the cutoff, mostly among men, and participation in KLPS-5 is about 8 percentage points lower among women above the cutoff. Both estimates are significant at the 10 percent level. Participation in at least one of KLPS-4 and KLPS-5, the rounds most relevant for the long-run outcomes, does not change at the cutoff for either gender. Completion of the KLPS-5 cognitive module is about 6 percentage points lower above the cutoff, a statistically insignificant difference driven by men. We return to this concern when discussing the cognitive results.

[10] Allowing the relationship between outcomes and KCPE scores to differ by gender on each side of the cutoff leaves the pooled estimates nearly unchanged, for example 0.162 rather than 0.157 for Form 4 completion and −100 rather than −105 dollars for non-farm earnings.

5 Results

5.1 Education Impacts

We begin by examining the first-stage impacts on educational attainment. Table 1 reports reduced-form estimates for completed years of schooling and key attainment thresholds. Attainment is measured for respondents interviewed in KLPS-2 or later, since respondents last observed in KLPS-1 were typically still enrolled and their attainment is censored before secondary schooling could be completed. Individuals scoring just above the KCPE cutoff completed, on average, 0.7 more years of schooling than those just below, significant at the 5 percent level. Consistent with this, the probability of attending any secondary school rises by 16.0 percentage points, while the likelihood of completing Form 4 increases by 15.7 percentage points (both significant at the 1 percent level). These are substantial effects relative to the control means, which indicate that only 57 percent of respondents in the below-cutoff sample ever attended secondary school, and 30 percent completed Form 4. Put differently, crossing the cutoff raises the Form 4 completion rate by more than half of its below-cutoff base. By contrast, we find no significant effect on progression beyond secondary school (13+ years), consistent with the admission rule operating on the secondary margin specifically.

Figure 1 visualizes the discontinuity in secondary school completion. The sharp jump at the cutoff confirms that KCPE scores generated meaningful quasi-experimental variation in secondary attainment. Figure 2 traces the discontinuity across attainment thresholds: the probability of attaining at least 9 through 12 years of schooling—the secondary grades—jumps by 13 to 16 percentage points at the cutoff, while the effect at 8 years is mechanically near zero (essentially everyone in the sample reached standard 8) and the effects at 13 or more years are small and statistically insignificant.

Appendix Table A5 reports results separately for men and women. The increases in attainment are concentrated among women, who gain 1.1 years of schooling and are 24.3

percentage points more likely to attend any secondary school and 22.0 percentage points more likely to complete Form 4. Among men the estimates are smaller and less precise: the increase in any secondary schooling is 10.8 percentage points, significant only at the 10 percent level, and the remaining estimates are positive but not statistically significant. This is consistent with the fact that the effective cutoff for women is lower (234 points versus 250 for men), meaning that girls just above the threshold were particularly likely to gain access to secondary school.

The KLPS cohort was also the study population for the Primary School Deworming Program, and prior work with these data finds that the long-run economic gains from childhood deworming are concentrated among males and among the older cohort—those over age 12 at the 1998 baseline (Hamory et al., 2021). Appendix Table A6 examines whether the schooling impacts at the KCPE cutoff load on these same cells, re-estimating the education outcomes separately by baseline cohort and sex. They do not: the attainment gains appear in both cohorts, no older–younger difference approaches statistical significance (p-values range from 0.37 to 0.92), and within each cohort the increases remain concentrated among women—the opposite of the male-concentrated deworming pattern. While the point estimates are somewhat larger in the older cohort, particularly for women, the overall stability across cohorts is reassuring that the first stage reflects the admission discontinuity itself rather than the cohort-by-sex heterogeneity that drives long-run deworming effects in this sample.

Together, these results replicate the first-stage findings in Ozier (2018) but with updated KLPS data. The magnitudes are comparable to earlier estimates, reinforcing the interpretation that admission policies induced substantial increases in attainment. Importantly, these effects motivate our focus on reduced-form outcomes: while admission strongly increases the probability of starting secondary school, it does not deterministically translate into additional years of completed schooling for all compliers. Admission itself may alter trajectories through expectations, peer groups, or parental investments, even absent full completion of secondary school. Our reduced-form strategy therefore captures the overall causal impact of

crossing the admission threshold.

5.2 Cognitive Impacts

We next examine the effect on adult cognitive performance. Table 2 reports estimates for domain-specific factor scores derived from the KLPS-5 cognitive battery of assessments (Gross et al., 2025), organized in three panels that move from association to causation. Panel A reports the raw gaps between respondents who completed secondary school and those who did not: completers outscore non-completers by 0.71 standard deviations in general cognition, with gaps of roughly 0.4 to 0.6 standard deviations in memory, language, executive function, and visuospatial ability and a smaller but still significant gap in orientation. These are the differences a cross-sectional comparison of adults would deliver, and they conflate any causal effect of secondary schooling with selection into it—pupils who completed secondary school had higher KCPE scores, and more advantaged backgrounds, than those who did not.

Panel B isolates the causal component using the regression discontinuity. Individuals just above the KCPE cutoff score higher on both general cognition and memory, with reduced-form gains of 0.23 and 0.28 standard deviations, respectively, both significant at the 5 percent level. For the other domains—language, executive function, orientation, and visuospatial ability—the point estimates are positive but not statistically significant. Two of these domains, orientation and visuospatial ability, exhibit pronounced ceiling effects in this comparatively young sample: the HCAP battery was designed for adults aged 60 and above, whereas our respondents average roughly 38 years of age, and most answer nearly all orientation items correctly. These measures therefore discriminate poorly at the top of the distribution, and the corresponding null results are less informative than those for the remaining domains.

Panel C rescales the reduced forms by the induced change in schooling, reporting fuzzy regression-discontinuity (IV) estimates of the effect of completing secondary school, instrumented with an indicator for scoring at or above the cutoff; each coefficient is the ratio

of the reduced-form and first-stage discontinuities at a common MSE-optimal bandwidth, interpretable as a local average treatment effect for compliers under the exclusion restriction discussed in Section 4. Completing secondary school is estimated to raise general cognition by 1.51 standard deviations (s.e. 0.80) and memory by 1.80 standard deviations (s.e. 0.91), both significant at the 5 percent level. These per-completion magnitudes are large but internally consistent with Panel B: compliers induced to complete Form 4 gain roughly five years of schooling (the 0.7-year attainment effect divided by the 0.16 completion first stage), so the implied gain is roughly 0.3 standard deviations per additional year of schooling—near the upper end of the range found for compulsory-schooling reforms in high-income countries (Banks and Mazzonna, 2012; Schneeweis et al., 2014)—and larger than, though statistically compatible with, the 0.6 standard deviation gain in combined vocabulary and reasoning performance that Ozier (2018) attributed to secondary school completion roughly a decade earlier in these respondents' lives. Because the first stage is only 0.16, however, the IV confidence intervals are wide—comfortably including the raw completer gaps of Panel A—so the comparison between Panels A and C is not sharp enough to quantify how much of the raw gap reflects selection.

Appendix Table A7 presents results separately by gender. Among men, we find a large and significant improvement in memory (0.36 SD, significant at the 5 percent level), while the estimated gain in general cognition (0.23 SD) is positive but not statistically significant at conventional levels. For women, the coefficient on the general cognitive score is similar in magnitude to men (0.21 SD), though not statistically significant at conventional levels; other domains are also not statistically significant, though the largest female point estimates appear for visuospatial ability (0.26 SD) and executive function (0.24 SD). One caveat applies to the male estimates in particular: completion of the C+ module is itself about 9 percentage points lower for men above the cutoff (Appendix Table A3), so the male results condition on a participation margin that is not fully balanced, while female participation is smooth. Taken together, these findings suggest that access to secondary schooling yields durable

improvements in adult cognitive performance.

5.3 Fertility Impacts

We then examine the effects on fertility. Table 3 reports reduced-form estimates for realized fertility (Panel A) and fertility intentions (Panel B). Adults just above the KCPE cutoff have substantially fewer children two decades later: total births fall by 0.39 (s.e. 0.256, significant at the 10 percent level), and births conditional on having any children decline by 0.56 (s.e. 0.244, significant at the 5 percent level). In contrast, extensive-margin fertility shows little change: the probability of having any children is essentially unchanged (0.023, s.e. 0.046), and age at first birth rises by 0.7 years (s.e. 0.60), neither statistically significant. Child mortality outcomes are also unchanged.

Fertility intentions are directionally consistent with these realized outcomes. Desired family size falls by 0.54 children in KLPS-2 (significant at the 1 percent level) and by 0.31 children in KLPS-3 (significant at the 5 percent level), indicating that admission to secondary school shifts preferences toward smaller families well before many births occur. The decline in desired fertility persists through KLPS-4 (-0.37, significant at the 5 percent level), by which point respondents were in their late 20s and early 30s.

Corresponding estimates by gender appear in Appendix Tables A8 (females) and A9 (males). Interestingly, the effects are larger for males relative to females: men just above the cutoff have 1.04 fewer children (significant at the 1 percent level), while there is no significant effect on fertility for female respondents (the point estimate is a statistically insignificant 0.22 additional children). This is especially striking given the schooling gains appeared stronger among female respondents than male respondents, the opposite pattern relative to fertility effects. There is a negative (though not statistically significant) point estimate for an indicator for teenage pregnancy for females of -0.056, relative to a control mean of 0.06. Desired fertility, by contrast, declines for both genders: the male estimates are significant in KLPS-2 and KLPS-4, and the female estimates are marginally significant in all three rounds.

Overall, the evidence points to a clear change in fertility trajectories—a reduction of roughly 0.4 children on average, and of roughly one child among men—accompanied by earlier-reported declines in desired family size. These patterns are broadly consistent with prior experimental and quasi-experimental evidence on the downstream demographic effects of secondary schooling in the region, though the patterns by gender differ, a contrast we return to in the conclusion.

Figure 3 plots reduced-form effects on cumulative births by age with 95% confidence intervals; the pooled effect becomes increasingly negative over adulthood. This is especially true for males (Panel C).

5.4 Marriage Impacts

One first-order explanation for changes in fertility patterns could be changes in marriage itself. Table 4 examines this channel directly, reporting marriage outcomes separately for men and women: marriage formation and timing (ever married, number of marriages, age at marriage), characteristics of the match (spousal age gap, spousal education and earnings), and marital structure and stability (polygamy, divorce, contraceptive use within marriage).

Marriage is nearly universal in this cohort—92 percent of below-cutoff men and 87 percent of below-cutoff women have ever married—and marriage formation is essentially unchanged at the cutoff. The effects on ever marrying (0.021 for men, 0.009 for women) and on the number of marriages are small and statistically insignificant, as are the effects on the probability of a polygamous union for men and of being a co-wife for women. Timing moves in the expected direction for women: age at first marriage rises by roughly 1.1 years, though the estimate is not statistically significant at conventional levels, consistent with the (also insignificant) delay in female age at first birth in Table 3 and with experimental evidence from Ghana that secondary education delays women's marriage and family formation (Duflo et al., 2026). Marital stability also shows little change: the only marginally significant estimate is a 5.4 percentage point increase in the probability that a man has ever divorced

(10 percent level), from a low base of 6 percent. Reported use of modern contraception with one's spouse is essentially unchanged for both men and women.

Turning to whom respondents marry, the results differ by gender. For women, spousal characteristics evolve smoothly through the cutoff: the education level, earnings, and age gap of the spouse are all statistically unchanged. For men, several estimates point toward matches with observably lower socioeconomic status: the education level of the first spouse falls by about 1.2 years (significant at the 5 percent level) against a below-cutoff mean of 9.5 years, and the spouse's earnings also decline (significant at the 10 percent level). Taken at face value, men induced toward secondary school do not match with more-educated spouses—if anything, the opposite—echoing the Ghanaian experimental finding that male scholarship winners tend to marry less-educated partners (Duflo et al., 2026). Given the number of marriage outcomes examined and the marginal significance of most of these estimates, we treat this pattern as suggestive rather than conclusive.

In sum, the fertility changes documented above occur within broadly stable marriage patterns: neither the likelihood, number, nor structure of marriages moves detectably at the cutoff, with a positive but imprecise delay in marriage for women and modest evidence of changed match characteristics for men.

5.5 Other Economic Outcomes

Table A10 reports reduced-form impacts at the KCPE cutoff on economic outcomes over the prior 12 months, pooling all available survey rounds. Crossing the cutoff has no detectable effect on total individual non-farm earnings or on per-capita household consumption. By component, farming profits decline (-$27.29, s.e. $16.89; significant at the 10% level)[11], while self-employment profits are negative but imprecise; wage earnings are positive yet imprecise. Total household earnings are positive but imprecisely estimated. These patterns suggest

[11]Measured in 2024 USD PPP. Estimates pool all available survey rounds (KLPS-2 through KLPS-5, with availability varying by outcome), include survey-round indicators and an indicator for female respondents as covariates, and cluster standard errors by respondent.

modest reallocation across income sources rather than large changes in total resources. The farming decline, while it registers statistically, is also small in absolute terms—below-cutoff farming profits average roughly $30 per year, against total non-farm earnings of roughly $2,200—so it is best read as a movement away from own-farm activity rather than a meaningful loss of resources.

Disaggregating by gender yields similar conclusions, though no individual component is statistically distinguishable from zero. Among men, farming profits fall (-$29.71, s.e. $23.52), though imprecisely, with other components near zero or imprecisely estimated; per-capita consumption is essentially unchanged. Among women, self-employment profits, wage earnings, total earnings, and per-capita consumption all remain close to zero and imprecise. Overall, crossing the KCPE cutoff appears to shift the composition of earnings slightly without clear effects on aggregate earnings or consumption. The direction of these shifts is consistent with the initial reduction in (largely low-skill) self-employment shown by Ozier (2018), though the long-run components are not precisely estimated. It also parallels the Ghanaian experimental evidence, where secondary scholarships produced no average earnings gains for men a decade later, and where women's gains came primarily through access to scarce public-sector jobs (Duflo et al., 2021, 2025).

5.6 Intergenerational Cognitive Effects

Table 5 reports RD estimates for the 8–14 year old children of KLPS respondents, separately by the gender of the KLPS respondent (not their child). For children of female respondents, the academic index rises by 0.36 standard deviations and the language index by 0.41 standard deviations, both significant at the 10 percent level, while the math estimate is positive but not statistically significant. There are no significant effects in the executive function and fluid intelligence domains. The probability that at least one child of a respondent was assessed evolves smoothly through the cutoff (Appendix Table A3), though because fertility itself responds to the cutoff—particularly for men—the set of children available for assessment

is partly an outcome, a caveat that applies especially to the male-side comparisons. These results are suggestive of potential pathways for intergenerational transmission, as female respondents had the largest increases in secondary school attainment, and this exposure may be more likely to translate into academic-related improvements for the next generation.

The mother-side concentration of these effects closely parallels the experimental evidence from Ghana: Duflo et al. (2026) find that secondary scholarships awarded to women—but not to men—improve the next generation's outcomes, including a 45 percent reduction in under-three mortality and cognitive gains of roughly 0.25 standard deviations once children reach school age. That the two settings, with different identifying variation, both concentrate intergenerational gains among the children of more-educated mothers is consistent with a long literature suggesting that maternal education matters especially for early-childhood investments (Breierova and Duflo, 2004).

6 Conclusion

The lasting changes in cognition that we document are novel, but in agreement with the work of Duflo et al. (2025) in Ghana. That study documents lasting increases in PISA-inspired test performance in math and reading comprehension. By contrast, the measures we include in this study are not based on these specific skills, but are instead based on general abilities drawn from the HCAP protocol; the largest cognitive effects we document are lasting benefits for memory. In instrumental-variables terms, completing secondary school raises general cognition among compliers by roughly 1.5 standard deviations—about 0.3 standard deviations per year of schooling (Table 2, Panel C). Together, the two studies suggest that the human capital produced by secondary school in Sub-Saharan Africa is real and durable, whether measured by academic skills a decade later or by general cognitive function two decades later—a finding relevant to the debate over how much effective instruction African secondary schools deliver.

We see very little evidence of increased earnings following secondary schooling. Our overall income estimates are relatively imprecise, but we see suggestive evidence—significant at the 10 percent level—of a decrease in the farm earnings of the individual who attended secondary school, consistent with earlier work by Ozier (2018). The lack of definitive changes in overall income is consistent with work in Ghana by Duflo et al. (2025), suggesting that in these lower middle income countries in Sub-Saharan Africa, secondary school alone may not be transformative in terms of income. The Ghanaian evidence also offers a candidate explanation for why average effects are modest: in a dual labor market where formal-sector jobs are rationed, the private payoff to secondary school arrives slowly and unevenly—in their setting, primarily among women who eventually secure scarce public-sector positions (Duflo et al., 2025). Our results, showing a shift out of own-farm activity without detectable gains elsewhere, are consistent with graduates repositioning in the labor market without (or not yet) realizing higher total earnings.

We find impacts of secondary schooling on desired fertility lasting many years after completing secondary school, with significant reductions in desired family size through the most recent survey round. The reduction in desired fertility contrasts with the findings of Brudevold-Newman (2021), who sees no impact on fertility intentions. One distinction may arise from the two study designs. The compliers in Brudevold-Newman's design experience a policy change that everyone around them also experiences, whereas in our regression discontinuity design, compliers gain an opportunity that their peers do not. Whether one sees oneself as following the same patterns as one's peers may influence outcomes including fertility intentions.

Consistent with Duflo et al. (2021) and Ozier (2018), we see marked and lasting reductions in fertility. Drawing on fertility history data and years of longitudinal surveys, we can see that the pattern could be consistent with a delay in fertility rather than a reduction in total fertility, but either could prove to be true in the years to come. The gender pattern, however, is essentially the reverse of the Ghanaian one: there, the fertility response is con-

centrated among women, who also delay marriage; here, realized fertility falls among men, while women—despite larger schooling gains at the cutoff—show delayed marriage but no detectable change in completed fertility. The marriage-market results indicate that the male fertility decline occurs without corresponding changes in whether, when, or how often men marry. These differences in fertility patterns by gender merit further exploration.

Though our study may not have the power to detect small changes in income, secondary school has changed the trajectory of students' lives in Kenya. Their human capital, measured by memory and other cognitive skills later in life, is raised two decades later. There is a shift away from farm income, and a demographic change in the timing and number of children, driven primarily by male respondents. The benefits appear to reach the next generation, concentrated among the children of women. In sum, we document a collection of longer-lasting changes than had ever been quantified before with quasi-experimental designs in Kenya, and we show that the human capital benefits of education can last a lifetime.

Table 1: Education Impacts

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Years of Schooling	10.10	0.738**	(0.320)	32.8	1,448	3,103
Any Secondary Schooling	0.57	0.160***	(0.054)	33.2	1,481	3,103
Completed Form 4	0.30	0.157***	(0.061)	36.4	1,613	3,103
13+ Years of Schooling	0.16	0.045	(0.056)	36.0	1,612	3,103

Notes: Educational attainment is measured from harmonized annual schooling histories in KLPS rounds 1–4 (1998–2021). *Years of Schooling* is the highest grade attained, with vocational training counted only for respondents who had completed Form 4; *Any Secondary Schooling*, *Completed Form 4*, and *13+ Years of Schooling* indicate attaining at least 9, 12, and 13 years. Attainment is defined only for respondents interviewed in KLPS-2 (2007–2009) or later; those last observed earlier were typically still enrolled. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 250 for males), so the coefficient on the discontinuity may be read directly. Estimates control for an indicator for female respondents (Section 4). Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table 2: Cognitive Impacts

Outcome	Comparison mean	Estimate	(SE)	Band-width	Eff. obs.	Obs.
Panel A: Secondary Completers versus Non-Completers (OLS)						
General Cognitive Factor Score	0.03	0.712***	(0.034)	–	–	2,418
Memory Factor Score	-0.03	0.595***	(0.038)	–	–	2,418
Language Factor Score	0.04	0.553***	(0.035)	–	–	2,418
Executive Function Factor Score	0.08	0.632***	(0.034)	–	–	2,418
Orientation Factor Score	0.06	0.162***	(0.035)	–	–	2,418
Visuospatial Factor Score	0.11	0.363***	(0.035)	–	–	2,418
Panel B: Reduced Form (Intent to Treat)						
General Cognitive Factor Score	0.04	0.226**	(0.122)	31.5	1,127	2,451
Memory Factor Score	-0.02	0.278**	(0.130)	34.8	1,225	2,451
Language Factor Score	0.07	0.028	(0.135)	32.0	1,128	2,451
Executive Function Factor Score	0.08	0.103	(0.124)	30.7	1,091	2,451
Orientation Factor Score	0.06	0.149	(0.137)	33.8	1,187	2,451
Visuospatial Factor Score	0.13	0.134	(0.132)	30.0	1,089	2,451
Panel C: Fuzzy RD (IV), Completing Secondary School						
General Cognitive Factor Score	0.04	1.508**	(0.795)	39.8	1,376	2,418
Memory Factor Score	-0.02	1.804**	(0.913)	36.9	1,291	2,418
Language Factor Score	0.07	0.217	(0.749)	37.7	1,319	2,418
Executive Function Factor Score	0.08	0.574	(0.719)	44.3	1,509	2,418
Orientation Factor Score	0.06	0.990	(0.873)	37.0	1,317	2,418
Visuospatial Factor Score	0.13	0.718	(0.798)	35.8	1,251	2,418

Notes: Cognitive outcomes are domain factor scores from the KLPS-5 C+ Module, which fielded the Harmonized Cognitive Assessment Protocol in Kenya (Gross et al., 2025): one score per domain, estimated by confirmatory factor analysis from a battery of more than 20 assessments and standardized to mean zero and unit variance in the full KLPS-5 sample, so means in the analysis sample need not equal zero. Panel A reports OLS coefficients on an indicator for completing secondary school, with heteroskedasticity-robust standard errors; the bandwidth and effective-observation columns do not apply (–). Panel B reports reduced-form RD estimates of scoring at or above the admission cutoff. Panel C reports fuzzy RD (IV) estimates in which completing secondary school is instrumented by an indicator for scoring at or above the cutoff, with the same kernel and bandwidth selection. *Comparison mean* is the mean among non-completers in Panel A and among respondents scoring below the cutoff in Panels B and C. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 250 for males), and RD estimates in Panels B and C control for an indicator for female respondents (Section 4). RD estimates are conventional local-linear coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points and *Eff. obs.* the observations within it. Standard errors (in parentheses) and significance stars in Panels B and C are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table 3: Fertility Impacts

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Panel A: Realized Fertility						
Indicator for any children	0.89	0.023	(0.046)	33.9	1,508	3,154
Total number of children	3.23	-0.392*	(0.256)	32.2	1,463	3,154
Age at first birth	23.71	0.673	(0.604)	35.3	1,364	2,669
Number of live births (cond> 0)	3.64	-0.561**	(0.244)	30.1	1,179	2,669
Under five mortality	0.05	0.003	(0.020)	38.6	3,956	7,242
Under one mortality	0.03	-0.002	(0.017)	34.0	4,318	8,862
Panel B: Fertility Intentions						
Desired No. Children (KLPS2)	3.38	-0.543***	(0.211)	23.3	919	2,550
Desired No. Children (KLPS3)	3.32	-0.307**	(0.146)	40.5	1,467	2,562
Desired No. Children (KLPS4)	4.06	-0.368**	(0.179)	38.9	1,361	2,449

Notes: Panel A reports realized fertility from retrospective birth histories in KLPS rounds 1–5: an indicator for any children, children ever born, age at first birth, live births among respondents with at least one child, and under-five and under-one mortality. The mortality outcomes are measured at the pregnancy level; their standard errors are clustered by respondent and their observation counts are pregnancies. Panel B reports the respondent's stated ideal number of children, including those already born, from KLPS-2, KLPS-3, and KLPS-4. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 250 for males), so the coefficient on the discontinuity may be read directly. Estimates control for an indicator for female respondents (Section 4). Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table 4: Marriage Impacts by Gender

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Among females:						
Ever married	0.87	0.009	(0.061)	27.4	547	1,344
Number of marriages	1.02	0.030	(0.097)	26.4	531	1,344
Age at marriage	22.10	1.056	(0.828)	35.1	604	1,143
Age gap with spouse	-5.41	0.946	(0.836)	24.3	428	1,138
Education level of spouse	10.18	0.193	(0.730)	34.6	580	1,131
Spouse's earnings	376.62	88.356	(94.835)	30.8	513	1,116
Modern birth control with spouse	0.66	-0.051	(0.099)	36.1	576	1,065
Ever a co-wife	0.26	0.075	(0.089)	31.4	534	1,139
Ever divorced	0.08	-0.034	(0.058)	34.9	584	1,139
Among males:						
Ever married	0.92	0.021	(0.057)	34.0	806	1,699
Number of marriages	1.33	-0.113	(0.150)	28.1	682	1,699
Age at marriage	25.00	0.469	(0.840)	31.3	661	1,477
Age gap with spouse	4.05	0.517	(0.572)	43.3	869	1,475
Education level of spouse	9.51	-1.224**	(0.561)	29.3	615	1,470
Spouse's earnings	160.15	-99.373*	(63.859)	39.9	795	1,449
Modern birth control with spouse	0.55	0.021	(0.099)	32.3	631	1,363
Ever polygamous	0.09	-0.069	(0.057)	33.6	539	1,148
Ever divorced	0.06	0.054*	(0.035)	38.9	793	1,476

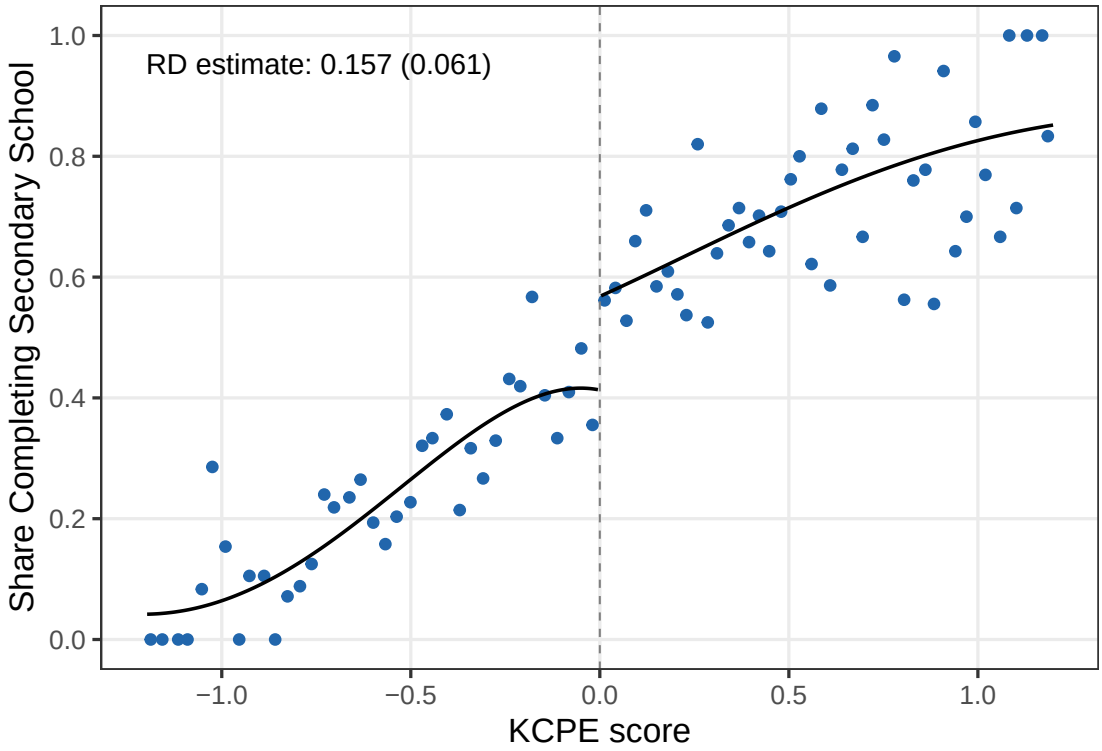
Notes: Marriage outcomes are constructed from the KLPS marriage histories. Age at marriage, age gap with spouse (the respondent's age at marriage minus the spouse's), spouse's education and earnings, and modern birth control with spouse describe the respondent's *first* marriage; ever married, number of marriages, ever polygamous / ever a co-wife, and ever divorced cover *all* marriages. Spouse's earnings are winsorized and expressed in 2024 USD PPP. *Ever polygamous* is male-reported and collected only in KLPS-5; *Ever a co-wife* is female-reported, collected from KLPS-2 onward. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 250 for males), so the coefficient on the discontinuity may be read directly. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table 5: Intergenerational Impacts by Parent's Gender

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Children of female respondents:						
Overall cognitive index	0.15	0.307	(0.215)	34.2	465	943
Academic index	0.16	0.362*	(0.212)	33.3	452	943
Language index	0.17	0.414*	(0.243)	31.0	410	943
Math index	0.13	0.245	(0.173)	42.2	581	943
Executive function index	0.10	0.215	(0.231)	35.4	492	943
Fluid intelligence index	0.09	-0.033	(0.276)	43.5	309	491
Strengths & difficulties index	0.11	0.105	(0.179)	38.6	533	945
Subjective health index	0.04	0.065	(0.195)	39.8	546	943
Home environment index	0.06	-0.044	(0.229)	32.6	433	941
Schooling investment index	0.09	0.129	(0.142)	31.4	423	945
Children of male respondents:						
Overall cognitive index	0.04	0.028	(0.228)	32.7	521	1,102
Academic index	0.04	-0.033	(0.219)	34.2	548	1,102
Language index	0.05	-0.009	(0.210)	37.5	604	1,102
Math index	0.02	-0.054	(0.224)	33.0	529	1,102
Executive function index	0.04	0.072	(0.242)	29.1	470	1,102
Fluid intelligence index	-0.01	-0.057	(0.333)	36.7	292	556
Strengths & difficulties index	0.03	-0.139	(0.252)	37.0	588	1,105
Subjective health index	0.04	-0.273	(0.244)	26.9	430	1,100
Home environment index	0.08	-0.001	(0.189)	37.4	605	1,104
Schooling investment index	-0.01	-0.102	(0.187)	31.1	504	1,106

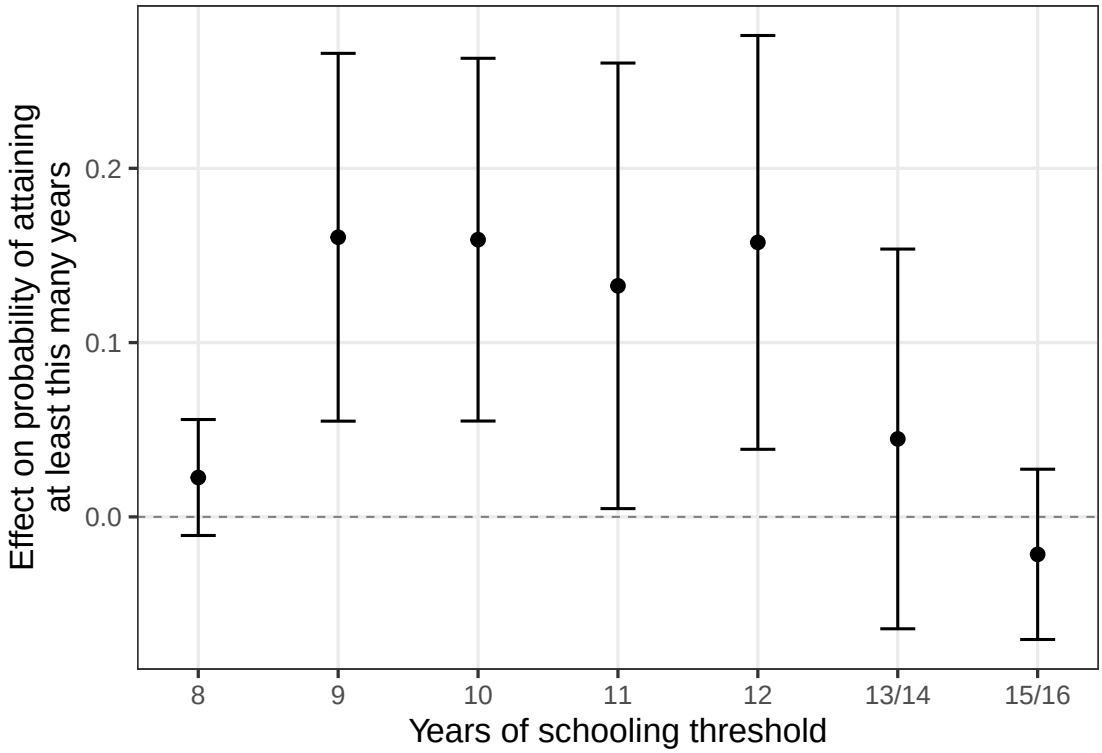
Notes: Outcomes are summary indices from the KLPS-5 child module, administered to up to two biological children aged 8–14 of each respondent; observations are children. The *Language* index combines the PPVT, animal naming, and Uwezo Swahili and English assessments; *Math* the Uwezo math assessment; *Executive function* digit span; and *Fluid intelligence* Raven's matrices, administered only to older children. The *Academic* index combines Language and Math, and the *Overall cognitive* index combines Language, Math, and Executive function. The *Strengths and difficulties* index combines the five SDQ subscales, with difficulties reverse-coded so that higher values are better; *Subjective health*, *Home environment*, and *Schooling investment* combine caregiver-reported items on the child's health, learning materials and activities in the home, and school enrollment, attendance, and spending. Each index is standardized to mean zero and unit variance in the full KLPS-5 child sample. Panels split the sample by the *parent's* sex, not the child's, with KCPE scores re-centered at the parent's gender-specific cutoff (234 for females, 250 for males), so the coefficient on the discontinuity may be read directly. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among children of parents scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Figure 1: Secondary School Completion



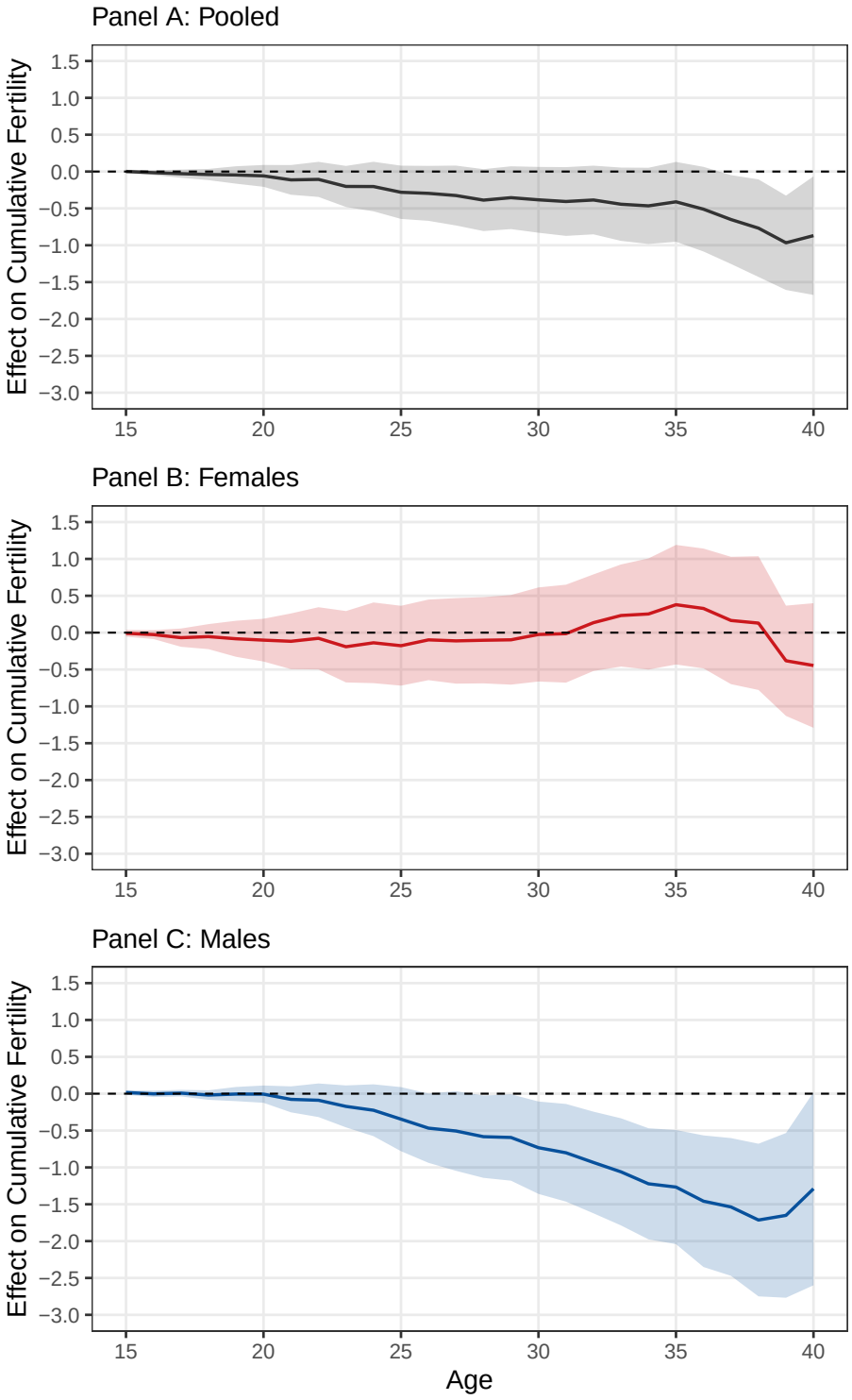
Notes: The figure plots the probability of completing Form 4 (the final year of secondary school) against the KCPE score, re-centered so that zero is the gender-specific cutoff. Completion is constructed from retrospective schooling histories in KLPS rounds 1–4 (see Table 1 for the education variables). Points are mean completion within evenly spaced `rdplot` bins of the running variable; the solid lines are local-linear fits estimated separately on each side of the cutoff. The printed estimate is the conventional local-linear RD coefficient at the cutoff, controlling for an indicator for female respondents, with its robust bias-corrected standard error in parentheses, as in Table 1.

Figure 2: Effects on Years-of-Schooling Thresholds



Notes: Each point is a separate conventional local-linear RD estimate, with its robust bias-corrected 95 percent confidence interval, of the effect of scoring at or above the cutoff on the probability of attaining at least the number of years of schooling on the horizontal axis, controlling for an indicator for female respondents. Years of schooling are the harmonized KLPS attainment measures, in the same sample as Table 1. The effect at 8 years is mechanically near zero, since everyone in the sample reached standard 8 to sit the KCPE. Attainment is recorded at 12, 14, and 16 years for completed secondary, some tertiary, and university, so no respondent attains exactly 13 or 15 years; the 13- and 14-year thresholds therefore define the same indicator, as do the 15- and 16-year thresholds, and each pair is shown as a single estimate (13/14 and 15/16).

Figure 3: Effect on Cumulative Fertility by Age



Notes: Each panel plots reduced-form RD estimates of the effect of scoring above the KCPE cutoff on cumulative fertility (children ever born by each age from 15 to 40), from retrospective birth histories in KLPS rounds 1–5. Panel A pools men and women, with scores re-centered at the gender-specific cutoff and a control for an indicator for female respondents; Panels B and C estimate separately for women and men (cutoffs of 234 and 250). Age is the calendar year minus the respondent's year of birth at baseline. The solid line is the conventional local-linear estimate age by age; shaded bands are 95 percent confidence intervals from the robust bias-corrected standard errors of `rdrobust` with MSE-optimal bandwidths. Negative values indicate fewer births among those just above the cutoff. Each age uses the respondents whose birth history extends to that age; roughly three-quarters of the sample contributes at age 36 and one-third at age 40, since the youngest cohorts have not yet reached the oldest ages.

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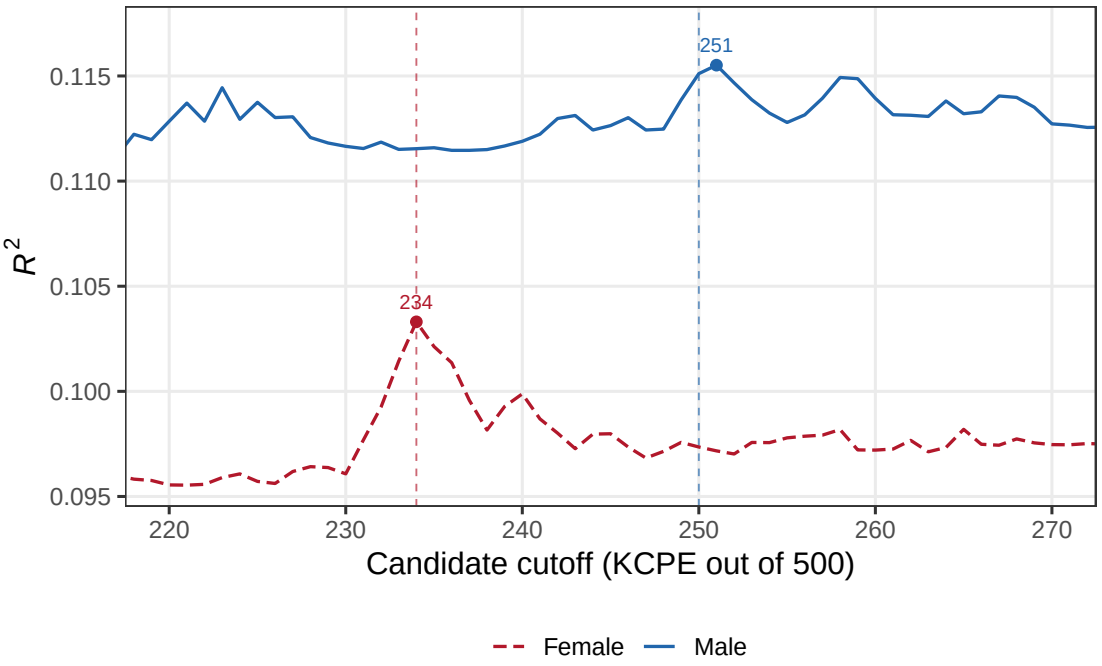
A Additional Tables and Figures

Table A1: Summary Statistics: KLPS Cohort and Matched Analysis Sample

	Full KLPS cohort	Matched analysis sample
Baseline characteristics		
Female	0.51 (0.50)	0.44 (0.50)
Year of birth	1985.4 (2.2)	1985.4 (2.2)
Age in 2024	38.6 (2.2)	38.6 (2.2)
Grade in 1998	4.48 (1.70)	4.78 (1.65)
Deworming treatment (early phase-in)	0.66 (0.47)	0.66 (0.47)
KCPE		
KCPE score (500-point scale)	–	244.6 (51.3)
Scored at or above cutoff	–	0.51 (0.50)
Survey participation		
Interviewed in KLPS-2 or later	0.88 (0.33)	0.98 (0.14)
Interviewed in KLPS-5 (to date)	0.71 (0.45)	0.77 (0.42)
Age at KLPS-5 interview	39.1 (2.3)	39.1 (2.2)
Adult outcomes		
Years of schooling	9.67 (3.06)	11.15 (2.63)
Any secondary schooling	0.53 (0.50)	0.72 (0.45)
Completed Form 4	0.32 (0.47)	0.50 (0.50)
Ever married	0.87 (0.33)	0.86 (0.35)
Number of marriages	1.15 (0.70)	1.10 (0.67)
Any children	0.86 (0.35)	0.85 (0.36)
Number of children	3.15 (2.16)	2.87 (2.00)
Lives in urban area (KLPS-4)	0.57 (0.49)	0.64 (0.48)
Total non-farm earnings, KLPS-5 (2024 USD PPP)	2,826 (4,061)	3,743 (4,770)
Any non-farm earnings, KLPS-5	0.78 (0.41)	0.81 (0.39)
Respondents	7,527	3,155

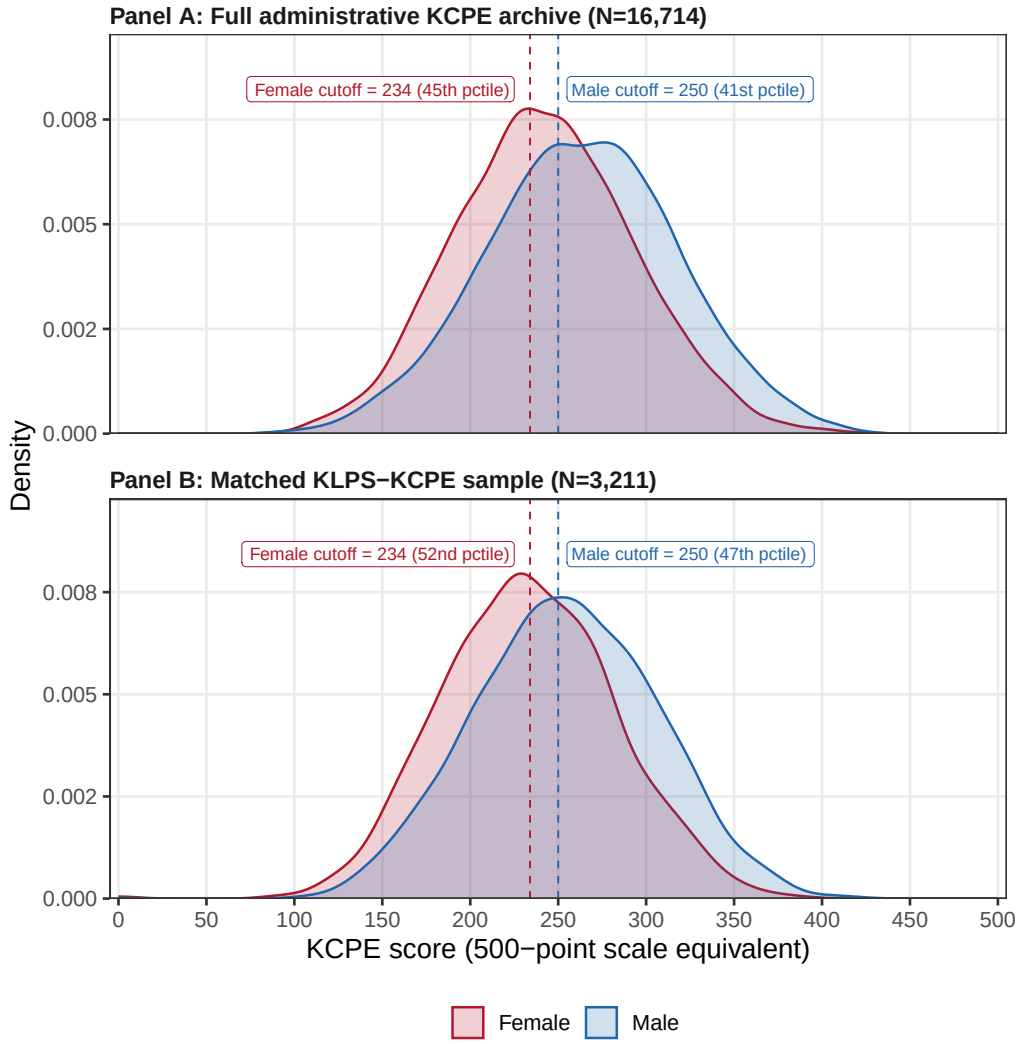
Notes: Means, with standard deviations in parentheses. The first column is the original KLPS cohort—pupils enrolled in grades 2–7 in the 75 Primary School Deworming Program schools in 1998; the second is the subset linked to a KCPE record that enters the analysis sample (Appendix B). *Female*, *Year of birth*, *Grade in 1998*, and *Deworming treatment* come from the 1998 baseline roster; year of birth is missing for about one-fifth of the cohort, and *Age in 2024* is 2024 minus year of birth. Linkage to a KCPE record requires having sat the examination, which accounts for the higher attainment, urbanization, and earnings of the matched sample. *KCPE score* is the first-attempt total on the 500-point scale (scores from the 1999–2000 examinations converted from 700 points), and *Scored at or above cutoff* refers to the gender-specific cutoffs of 234 for females and 250 for males. *Age at KLPS-5 interview* is the interview year minus year of birth among respondents interviewed to date. Schooling outcomes are defined for respondents interviewed in KLPS-2 or later, as in Table 1; marriage and fertility outcomes follow Tables 4 and 3; *Lives in urban area* is measured at the KLPS-4 interview, the most recent round with a harmonized urban indicator; earnings are from completed KLPS-5 interviews, in 2024 USD PPP, trimmed of the top one percent. Each row uses the respondents for whom the variable is observed.

Figure A1: Structural Break Search



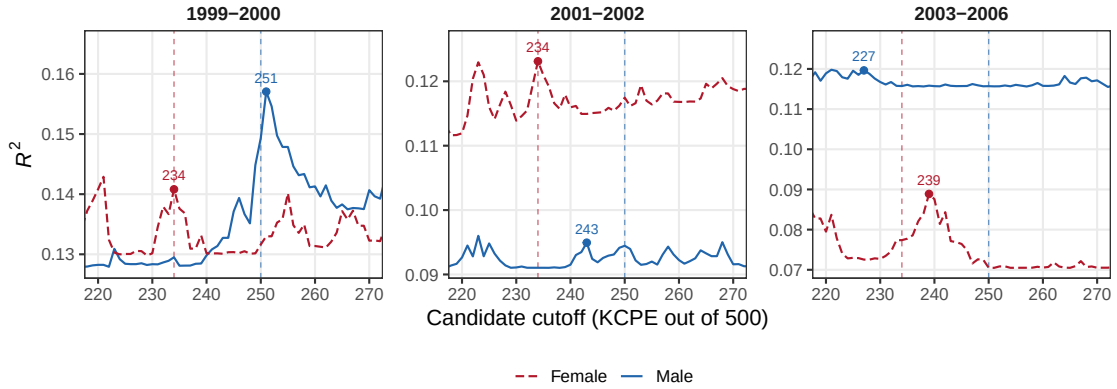
Notes: The figure implements a structural-break search in the spirit of Card et al. (2008) for the KCPE score at which the probability of completing secondary school (Form 4) rises discontinuously. For each candidate cutoff between 200 and 300 on the 500-point scale we fit a piecewise-linear model for completion within a symmetric ±70-point window and plot its fit (R^2), separately for females and males; scores from the pre-2001 700-point examinations are converted to 500-point equivalents. The figure displays the 220–270 range, where the fits peak; the R^2 profiles decline outside it. The female series peaks at 234 and the male series at 251, marked by the vertical lines; we adopt cutoffs of 234 for females and 250 for males—the latter for institutional comparability, and because it is more stable across subperiods. Appendix Figure A3 repeats the search separately by exam-year group.

Figure A2: KCPE Score Distributions by Gender



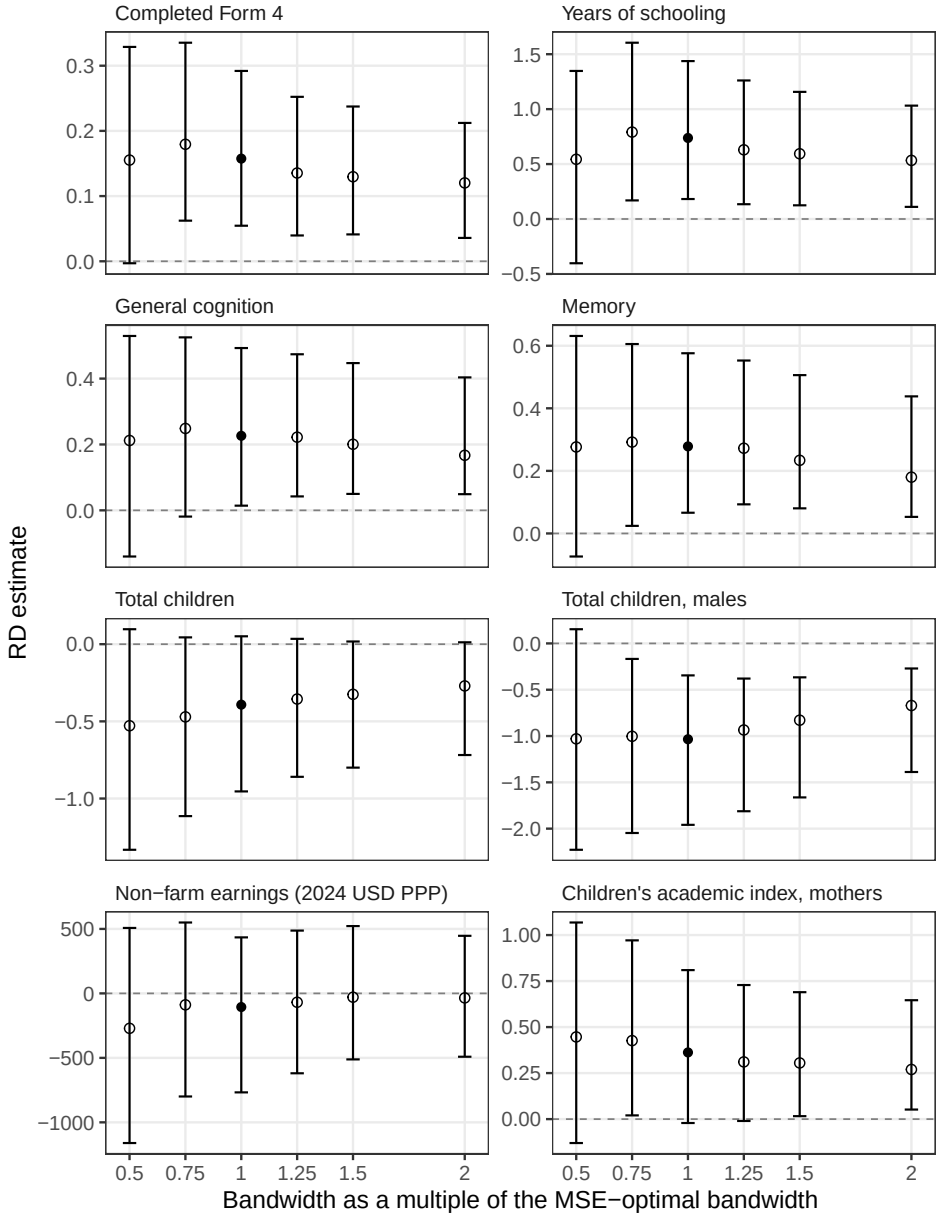
Notes: The figure plots kernel densities of KCPE scores for female and male examinees, in the full administrative KCPE archive (Panel A) and in the matched KLPS–KCPE sample (Panel B). Scores from the 1999–2000 examinations, graded out of 700 points, are converted to 500-point equivalents, so all scores are shown on the 500-point scale. Panel A includes the 16,714 archive records from examination years 1999–2006 with a valid score and recorded sex (of 17,384 records in the archive overall); Panel B includes the 3,211 matched KLPS respondents with a verified KCPE score and exam year in the same range, before the cohort and exam-timing restrictions that define the analysis sample. Dashed vertical lines mark the gender-specific admission cutoffs (234 for females, 250 for males), and each label reports the percentile of the relevant gender-specific score distribution at which that cutoff falls. Both cutoffs fall near the middle of the distribution, where the data are dense, and the matched sample closely tracks the archive from which it is drawn.

Figure A3: Structural Break Search by KCPE Year Group



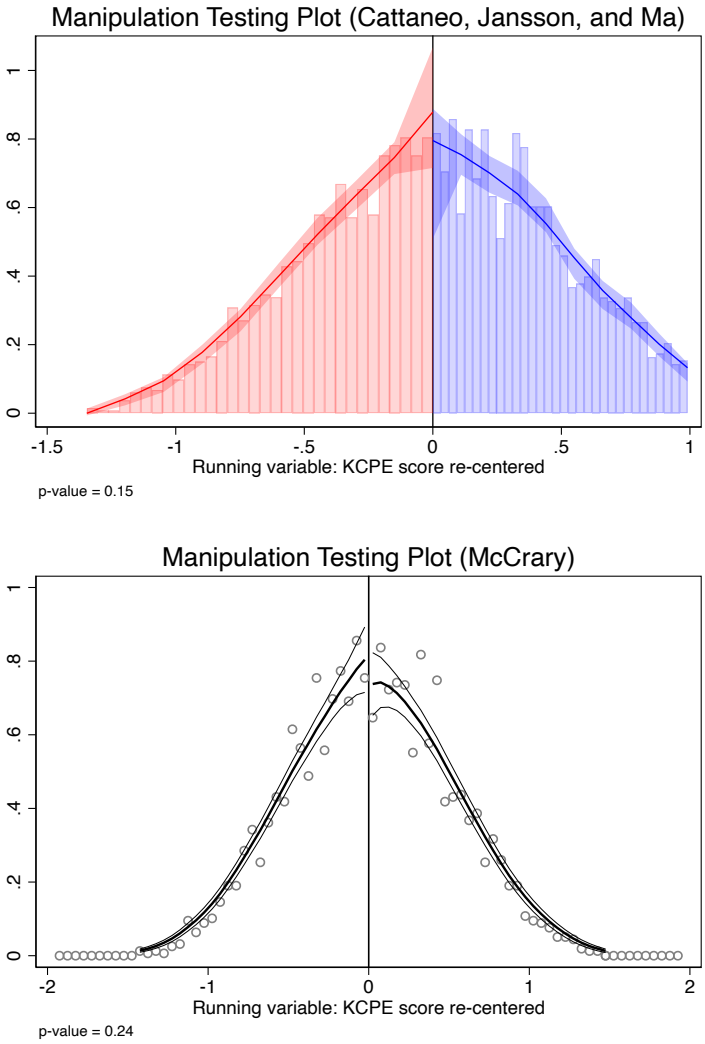
Notes: This figure repeats the structural-break search of Appendix Figure A1 separately by KCPE examination year group (1999–2000, 2001–2002, and 2003–2006) and by gender. For each candidate cutoff between 200 and 300 on the 500-point scale, we fit the same piecewise-linear model for Form 4 completion within a symmetric ±70-point window in each gender-by-year-group cell and plot the resulting R^2; as in Appendix Figure A1, the figure displays the 220–270 range. Scores from 1999–2000 are converted to 500-point equivalents, and the 2003–2006 group folds in the small number of examinees who sat the KCPE in 2006. Dashed vertical lines mark the pooled gender-specific cutoffs (234 for females, 250 for males); dots mark each cell's R^2-maximizing candidate between 225 and 275 (among candidates implying a positive discontinuity). Each cell contains roughly 280–620 observations within the search window, versus roughly 1,100–1,400 pooled, so the subperiod profiles are flatter and noisier. The female search peaks at exactly 234 in both 1999–2000 and 2001–2002 and at 239 in 2003–2006; the male search peaks at 251 in 1999–2000, matching the pooled search, while the flatter 2001–2002 and 2003–2006 male profiles peak at 243 and 227. In three of the six cells the full-range maximizer lies outside this interior range (at 213, 223, and 298), but each sits in a nearly flat profile, exceeding the interior peak by at most 0.012: those subsamples carry little information about the precise threshold location, in contrast with the sharply peaked pooled search.

Figure A4: Sensitivity of the Main Estimates to the Bandwidth



Notes: Each point is a conventional local-linear RD estimate using a bandwidth equal to the indicated multiple of the outcome's MSE-optimal bandwidth, with the bias-correction bandwidth scaled by the same multiple. Bars are robust bias-corrected 95 percent confidence intervals (Calonico et al., 2014). Filled points use the MSE-optimal bandwidth and reproduce the estimates in Tables 1, 2, 3, and 5 and Appendix Tables A9 and A10. The MSE-optimal bandwidths range from 26 to 36 KCPE points. Pooled estimates control for an indicator for female respondents. Non-farm earnings pool KLPS-2 through KLPS-5, include survey-round indicators, and cluster standard errors by respondent. The children's academic index is for children of female respondents.

Figure A5: Density test



Notes: The figure reports manipulation tests for the running variable—the KCPE score in the matched analysis sample (N=3,162), re-centered at the gender-specific cutoff (234 for females, 250 for males) and pooled across genders. The top panel is the local-polynomial density test of Cattaneo et al. (2020) (`rddensity`); the bottom panel is the McCrary (2008) discontinuity test (`DCdensity`). Both fail to reject continuity of the density at the cutoff (p-values reported in each panel), consistent with no sorting in the test score around the threshold. Local-polynomial density tests run separately by gender also fail to reject ($p = 0.92$ for females and $p = 0.74$ for males), and Appendix Table A4 reports an archive-side test of whether record linkage is smooth through the cutoff.

Table A2: Covariate Balance at the KCPE Cutoff

Outcome	Below-cutoff mean	RD estimate		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Pooled:						
Female	0.47	-0.065*	(0.043)	33.3	1,508	3,162
Year of birth	1984.88	0.212	(0.280)	28.0	1,191	2,938
Grade in 1998	4.99	0.060	(0.169)	41.2	1,832	3,162
Father's education	9.34	-0.774	(0.824)	27.0	1,150	2,940
Mother's education	6.06	0.226	(0.520)	37.7	1,587	2,966
Deworming Treatment	0.66	-0.033	(0.054)	43.3	1,899	3,155
School average test score, 1996	-0.02	0.022	(0.076)	40.8	1,799	3,155
Pupils at baseline school, 1998	397.53	29.943	(54.826)	40.3	1,795	3,155
Among females:						
Year of birth	1985.19	-0.024	(0.380)	30.4	594	1,292
Grade in 1998	5.03	0.253	(0.269)	33.4	690	1,397
Father's education	9.75	-0.025	(1.128)	30.4	585	1,289
Mother's education	6.19	0.574	(0.771)	34.2	662	1,305
Deworming Treatment	0.67	0.004	(0.089)	29.9	618	1,390
School average test score, 1996	-0.03	-0.050	(0.085)	29.9	618	1,390
Pupils at baseline school, 1998	396.21	53.022	(58.474)	35.6	733	1,390
Among males:						
Year of birth	1984.61	0.394	(0.412)	29.1	679	1,646
Grade in 1998	4.95	-0.177	(0.272)	33.5	820	1,765
Father's education	8.99	-1.277	(1.198)	26.0	624	1,651
Mother's education	5.94	-0.179	(0.800)	32.2	749	1,661
Deworming Treatment	0.66	-0.015	(0.075)	44.6	1,052	1,765
School average test score, 1996	-0.01	0.074	(0.091)	50.6	1,163	1,765
Pupils at baseline school, 1998	398.68	14.745	(57.444)	43.4	1,027	1,765

Notes: The table reports RD estimates for characteristics determined before the KCPE examination, in the matched analysis sample. *Female*, *Year of birth*, and *Grade in 1998* come from the 1998 baseline pupil roster. *Father's education* and *Mother's education* are the highest levels attained, in years, as reported in KLPS-2 (as in Ozier 2018), filled where missing from the corresponding KLPS-3 and then KLPS-4 questions converted to the same years scale; together these cover about 93 percent of the matched sample. *Deworming Treatment* indicates assignment to the first two phase-in groups (treated from 1998–99) of the Primary School Deworming Program (Miguel and Kremer, 2004). *School average test score, 1996* (in standard-deviation units) and *Pupils at baseline school, 1998* describe the baseline primary school; these two rows cluster standard errors by school. The *Female* row appears only in the pooled panel, since pooling stacks the two gender-specific cutoffs. The other pooled rows control for an indicator for female respondents (Section 4). KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 250 for males), so the coefficient on the discontinuity may be read directly. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A3: Survey Participation at the KCPE Cutoff

Outcome	Below-cutoff mean	RD estimate		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Pooled:						
Interviewed in KLPS-2 or later	0.98	-0.033*	(0.023)	32.9	1,476	3,162
Interviewed in KLPS-2	0.82	-0.066*	(0.045)	34.4	1,555	3,162
Interviewed in KLPS-3	0.83	-0.008	(0.051)	37.1	1,690	3,162
Interviewed in KLPS-4	0.77	0.021	(0.048)	44.0	1,938	3,162
Interviewed in KLPS-5 (to date)	0.90	-0.030	(0.035)	42.1	1,861	3,162
Interviewed in KLPS-4 or KLPS-5	0.94	-0.013	(0.026)	38.8	1,731	3,162
Completed C+ cognitive module	0.82	-0.062	(0.049)	37.3	1,692	3,162
Any child assessed, ages 8–14	0.56	-0.021	(0.062)	36.4	1,644	3,162
Among females:						
Interviewed in KLPS-2 or later	0.98	-0.019	(0.026)	36.9	761	1,397
Interviewed in KLPS-2	0.82	-0.060	(0.069)	34.0	706	1,397
Interviewed in KLPS-3	0.81	0.022	(0.085)	32.3	665	1,397
Interviewed in KLPS-4	0.79	0.079	(0.073)	33.4	690	1,397
Interviewed in KLPS-5 (to date)	0.93	-0.078*	(0.056)	30.6	638	1,397
Interviewed in KLPS-4 or KLPS-5	0.95	0.000	(0.040)	29.7	620	1,397
Completed C+ cognitive module	0.84	-0.020	(0.073)	30.4	638	1,397
Any child assessed, ages 8–14	0.57	0.059	(0.099)	28.5	588	1,397
Among males:						
Interviewed in KLPS-2 or later	0.98	-0.045	(0.036)	32.7	805	1,765
Interviewed in KLPS-2	0.82	-0.117*	(0.081)	21.3	566	1,765
Interviewed in KLPS-3	0.86	-0.026	(0.060)	42.8	1,005	1,765
Interviewed in KLPS-4	0.75	-0.008	(0.065)	45.0	1,052	1,765
Interviewed in KLPS-5 (to date)	0.88	0.003	(0.054)	35.6	869	1,765
Interviewed in KLPS-4 or KLPS-5	0.93	-0.027	(0.042)	33.7	825	1,765
Completed C+ cognitive module	0.80	-0.093	(0.066)	39.6	955	1,765
Any child assessed, ages 8–14	0.55	-0.072	(0.089)	33.6	820	1,765

Notes: The table reports RD estimates for survey participation in the matched analysis sample. *Interviewed in KLPS-2 or later* is the measurement rule for educational attainment (Table 1). The round-specific rows indicate completion of a main interview in that round; KLPS-5 is ongoing, so its row reflects interviews completed to date, and *Interviewed in KLPS-4 or KLPS-5* combines the two most recent rounds. *Completed C+ cognitive module* indicates completion of the KLPS-5 cognitive assessment (Table 2); *Any child assessed, ages 8–14* indicates that at least one biological child of the respondent completed the child assessment module. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 250 for males), so the coefficient on the discontinuity may be read directly. Pooled estimates control for an indicator for female respondents (Section 4). Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A4: KCPE Archive Match Rates at the Cutoff

Outcome	Below-cutoff mean	RD estimate		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
All examinees:						
Matched to a KLPS respondent	0.22	-0.026	(0.021)	38.1	8,649	16,714
In the analysis sample	0.22	-0.023	(0.021)	37.0	8,452	16,714
Female examinees:						
Matched to a KLPS respondent	0.23	-0.056**	(0.032)	36.7	3,718	7,153
In the analysis sample	0.23	-0.049*	(0.032)	36.0	3,621	7,153
Male examinees:						
Matched to a KLPS respondent	0.22	-0.001	(0.028)	41.5	5,089	9,561
In the analysis sample	0.21	-0.001	(0.028)	40.4	4,972	9,561

Notes: The table reports RD estimates, over KCPE archive records, of the probability that a record is linked to the KLPS—a test of whether linkage is smooth through the cutoff. The sample is the 16,714 archive records from examination years 1999–2006 with a valid score and recorded sex; scores from 1999–2000 are converted to 500-point equivalents, and the panels split records by the sex recorded in the archive. *Matched to a KLPS respondent* indicates linkage by the procedure of Appendix B; *In the analysis sample* additionally requires that the linked respondent enters the analysis sample: of 3,259 accepted matches, 58 are dropped because the exam year falls more than one year before or three years after the year implied by the respondent's baseline grade and 39 because the matched record lacks a recorded score, leaving 3,162. Match rates are low because the archive also contains cohorts outside the KLPS sampling frame and repeat attempts: at most 4,948 records—one per KLPS respondent who reached standard 8—could link, so the maximum attainable rate is 0.30, and the realized below-cutoff rate is about three-quarters of it. The female match rate falls by 5–6 percentage points at the cutoff; Section 4 and Appendix C present evidence that this reflects repeat examination-taking rather than differential linkage of respondents, and that the density of matched respondents and covariate balance are smooth at the cutoff for both genders (Appendix Figure A5; Appendix Table A2). KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 250 for males), so the coefficient on the discontinuity may be read directly. Estimates for all examinees control for an indicator for female examinees (Section 4). Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among records scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A5: Education Impacts by Gender

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Among females:						
Years of Schooling	9.78	1.061**	(0.491)	28.5	577	1,369
Any Secondary Schooling	0.49	0.243***	(0.097)	24.7	521	1,369
Completed Form 4	0.25	0.220***	(0.094)	30.4	627	1,369
13+ Years of Schooling	0.13	0.051	(0.077)	36.6	742	1,369
Among males:						
Years of Schooling	10.38	0.489	(0.435)	34.5	834	1,734
Any Secondary Schooling	0.63	0.108*	(0.062)	44.6	1,032	1,734
Completed Form 4	0.35	0.113	(0.086)	37.1	899	1,734
13+ Years of Schooling	0.18	0.044	(0.081)	32.1	785	1,734

Notes: The table re-estimates the Table 1 education outcomes separately for females and males; see Table 1 for construction. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 250 for males), so the coefficient on the discontinuity may be read directly. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A6: Education Impacts by Baseline Cohort and Sex

Outcome	Older cohort (age >12 in 1998)		Younger cohort (age ≤12 in 1998)		Difference	
	Below-cutoff mean	Estimate	Below-cutoff mean	Estimate	Older − younger	p-value
Panel A: Full Sample						
Years of Schooling	9.82	0.738*	10.58	0.447	0.292	0.69
		(0.404)		(0.611)	(0.733)	
Any Secondary Schooling	0.51	0.159**	0.66	0.136	0.022	0.86
		(0.069)		(0.108)	(0.128)	
Completed Form 4	0.25	0.185**	0.38	0.100	0.085	0.54
		(0.086)		(0.107)	(0.137)	
13+ Years of Schooling	0.12	0.059	0.22	-0.011	0.070	0.57
		(0.077)		(0.095)	(0.122)	
Observations	1,517		1,365			
Panel B: Females						
Years of Schooling	9.46	1.557**	10.21	1.035	0.522	0.65
		(0.708)		(0.895)	(1.141)	
Any Secondary Schooling	0.43	0.225**	0.58	0.246*	-0.021	0.92
		(0.117)		(0.167)	(0.204)	
Completed Form 4	0.19	0.399***	0.32	0.202	0.197	0.37
		(0.157)		(0.154)	(0.220)	
13+ Years of Schooling	0.10	0.126	0.18	0.034	0.092	0.61
		(0.136)		(0.117)	(0.179)	
Observations	626		641			
Panel C: Males						
Years of Schooling	10.10	0.534	11.00	-0.210	0.744	0.43
		(0.518)		(0.784)	(0.939)	
Any Secondary Schooling	0.57	0.131	0.74	0.034	0.096	0.46
		(0.081)		(0.103)	(0.131)	
Completed Form 4	0.30	0.147	0.45	-0.017	0.164	0.37
		(0.116)		(0.142)	(0.183)	
13+ Years of Schooling	0.14	0.063	0.26	-0.058	0.121	0.51
		(0.097)		(0.154)	(0.182)	
Observations	891		724			

Notes: The table re-estimates the Table 1 education outcomes separately by baseline cohort and sex; see Table 1 for construction. Cohorts follow Hamory et al. (2021): respondents over age 12 at the 1998 baseline (older) versus age 12 or under (younger), with age and sex from the baseline sample master; 221 respondents lacking a baseline birth year are dropped. Each cell is a separate conventional local-linear RD estimate from `rdrobust` with a triangular kernel and its own MSE-optimal bandwidth, KCPE scores re-centered at the gender-specific cutoffs; Panel A estimates control for an indicator for female respondents (Section 4); robust bias-corrected standard errors are in parentheses, and *Below-cutoff mean* is the outcome mean among below-cutoff respondents in the relevant subsample. *Older − younger* is the difference in the two cohort estimates, with standard error $\sqrt{\text{se}^2_{\text{older}} + \text{se}^2_{\text{younger}}}$ since the subsamples are disjoint; the p-value tests equality. Panels B and C partition Panel A. * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A7: Cognitive Impacts by Gender

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Among females:						
General Cognitive Factor Score	-0.03	0.210	(0.161)	38.6	636	1,108
Memory Factor Score	-0.02	0.189	(0.191)	33.7	547	1,108
Language Factor Score	0.03	-0.105	(0.201)	31.8	520	1,108
Executive Function Factor Score	-0.06	0.241	(0.191)	30.2	505	1,108
Orientation Factor Score	-0.01	0.146	(0.201)	38.7	636	1,108
Visuospatial Factor Score	0.00	0.261	(0.191)	32.4	528	1,108
Among males:						
General Cognitive Factor Score	0.11	0.225	(0.166)	34.2	654	1,343
Memory Factor Score	-0.03	0.355**	(0.193)	33.9	640	1,343
Language Factor Score	0.11	0.132	(0.176)	34.8	660	1,343
Executive Function Factor Score	0.21	-0.033	(0.161)	28.8	548	1,343
Orientation Factor Score	0.12	0.168	(0.180)	32.0	620	1,343
Visuospatial Factor Score	0.25	0.002	(0.172)	31.2	605	1,343

Notes: The table re-estimates the reduced-form (Panel B) cognitive outcomes of Table 2 separately for females and males; see Table 2 for the factor-score construction. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 250 for males), so the coefficient on the discontinuity may be read directly. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A8: Fertility Impacts: Females

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Realized Fertility:						
Indicator for any children	0.89	0.075	(0.072)	29.5	617	1,395
Total number of children	3.14	0.216	(0.395)	29.0	591	1,395
Age at first birth	22.37	0.591	(0.952)	32.0	560	1,200
Teenage pregnancy	0.06	-0.056	(0.051)	31.4	651	1,395
Number of live births (cond> 0)	3.53	-0.038	(0.364)	28.7	503	1,200
Under five mortality	0.06	0.015	(0.035)	29.9	1,418	3,332
Under one mortality	0.04	-0.004	(0.028)	33.5	1,869	3,892
Fertility Intentions:						
Desired No. Children (KLPS2)	3.26	-0.443*	(0.301)	26.4	457	1,125
Desired No. Children (KLPS3)	3.24	-0.390**	(0.205)	47.6	752	1,096
Desired No. Children (KLPS4)	3.85	-0.404*	(0.283)	28.4	471	1,097

Notes: The table re-estimates the Table 3 fertility outcomes for women, adding an indicator for teenage pregnancy (a first birth before age 18, zero for respondents with no children); see Table 3 for construction. Mortality outcomes are measured at the pregnancy level, with standard errors clustered by respondent. KCPE scores are re-centered at the female cutoff of 234. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A9: Fertility Impacts: Males

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Realized Fertility:						
Indicator for any children	0.88	-0.017	(0.067)	30.5	751	1,759
Total number of children	3.31	-1.035***	(0.412)	25.8	649	1,759
Age at first birth	24.90	0.636	(0.779)	38.8	791	1,469
Teenage pregnancy	0.03	0.009	(0.019)	30.4	751	1,759
Number of live births (cond> 0)	3.75	-1.082***	(0.346)	28.7	599	1,469
Under five mortality	0.05	-0.003	(0.025)	41.5	2,216	3,910
Under one mortality	0.02	0.002	(0.018)	41.1	2,819	4,970
Fertility Intentions:						
Desired No. Children (KLPS2)	3.48	-0.515***	(0.236)	33.3	660	1,425
Desired No. Children (KLPS3)	3.40	-0.217	(0.203)	36.7	744	1,466
Desired No. Children (KLPS4)	4.25	-0.496*	(0.315)	29.5	565	1,352

Notes: The table re-estimates the Table 3 fertility outcomes for men, adding an indicator for teenage pregnancy (a first birth before age 18, zero for respondents with no children); see Table 3 for construction. Mortality outcomes are measured at the pregnancy level, with standard errors clustered by respondent. KCPE scores are re-centered at the male cutoff of 250. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A10: Earnings and Consumption Impacts

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Pooled:						
Total non-farm earnings, 12m	2187.02	-105.184	(327.051)	32.3	4,612	9,761
Farming profits, 12m	30.05	-27.292*	(16.886)	29.5	2,144	4,914
Self-employment profits, 12m	532.72	-56.393	(106.327)	34.6	4,893	9,814
Wage earnings, 12m	1504.90	148.778	(277.549)	34.5	4,870	9,745
Per capita household consumption, 12m	3598.07	30.854	(494.471)	30.5	1,222	2,687
Total household earnings, 12m	1904.89	377.282	(347.928)	27.4	1,913	4,672
Among females:						
Total non-farm earnings, 12m	1151.05	13.876	(370.089)	29.6	1,983	4,364
Farming profits, 12m	25.73	-16.855	(18.379)	35.3	1,155	2,154
Self-employment profits, 12m	321.04	-4.769	(111.876)	36.6	2,403	4,369
Wage earnings, 12m	753.60	51.154	(305.734)	37.7	2,473	4,360
Per capita household consumption, 12m	2996.00	-34.969	(484.323)	29.2	552	1,217
Total household earnings, 12m	1473.14	432.564	(339.135)	36.0	1,171	2,133
Among males:						
Total non-farm earnings, 12m	3119.78	-214.135	(546.712)	30.9	2,346	5,397
Farming profits, 12m	33.81	-29.706	(23.519)	35.7	1,383	2,760
Self-employment profits, 12m	722.91	-91.692	(175.426)	34.0	2,573	5,445
Wage earnings, 12m	2180.84	275.293	(476.804)	30.4	2,337	5,385
Per capita household consumption, 12m	4170.51	61.254	(862.797)	29.5	627	1,470
Total household earnings, 12m	2307.57	447.833	(603.247)	26.0	981	2,539

Notes: Monetary outcomes are expressed in 2024 USD PPP and trimmed of the top one percent within each survey round. All outcomes cover the last 12 months. Estimates pool observations across KLPS rounds, with availability varying by outcome (total non-farm, self-employment, and wage earnings in KLPS-2 through KLPS-5; farming profits and per-capita consumption in KLPS-3 and KLPS-4; household earnings in KLPS-4 and KLPS-5), include survey-round indicators as covariates (and, in the pooled panel, an indicator for female respondents), and cluster standard errors by respondent. *Total non-farm earnings* is the sum of wage earnings across all jobs and non-agricultural self-employment profits across all businesses. *Farming profits* is net revenue from farming for which the respondent supplied the household labor and was the main decision-maker, less input costs. *Per capita household consumption* is the value of food and non-food consumption divided by household size, with an urban price adjustment for Nairobi and Mombasa. *Total household earnings* is the sum of wage earnings, self-employment profits, and agricultural profits across household members, divided by household size. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 250 for males). Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

B Matching Approach

This appendix describes the procedure used to link individual records from the Kenya Life Panel Survey (KLPS) to the Kenya Certificate of Primary Education (KCPE) administrative data. The KLPS dataset contains detailed survey information on pupils, including multiple reported names across survey rounds and records of the primary schools they attended. The KCPE data contain the official examination records with candidate names, schools, years, and exam scores. We follow a strict matching procedure with the goal of linking each KCPE record to a single KLPS respondent, based on consistency between reported names and schools.

Step 1: Assembling identifying information. We began by compiling all available name and school variables from the KLPS surveys. In each KLPS round, respondents are asked to report their name again, and across rounds they often provide slightly different versions or spellings. We take advantage of all names ever reported in KLPS to maximize the chances of identifying a correct match in the KCPE records. In addition, KLPS collects information on schooling history, including the primary schools attended, which we draw on alongside names. On the KCPE side, we use the candidate's full name, school in which they sat for the exam, and the year of examination. By working with the full history of reported names and schools, we aimed to capture as much identifying information as possible, recognizing that spelling or reporting could vary over time.

Step 2: Cleaning and standardizing names. All names were first converted to uppercase and stripped of punctuation and extraneous whitespace. Beyond these generic edits, we created specific replacement rules for common variants, such as standardizing "Maureen/Morine/Maurine" to MAUREEN, "Bernard/Benard" to BERNARD, and "Jackline/Jacqueline" to JACKLINE. We also harmonized common regional variants of surnames, for example treating ODUOR, ODUAR, and ODOUR as the same name standardized to ODUOR. Each name was then broken into tokens (for example, "John Paul Otieno" became {John, Paul, Otieno}), allowing us to match across different name orders and combinations.

Step 3: Cleaning and standardizing schools. A similar cleaning process was applied to school names. We removed suffixes like "Primary School," "PS," or "PRI SCH," and corrected frequent transcription mistakes. In cases where the same school was referred to with multiple spellings or abbreviations, these were collapsed into a single standardized form.

Step 4: Restricting matches by school. The first filter required that survey data and KCPE records agree on school. Using the cleaned and standardized school names, we matched each survey school to the KCPE schools exactly, blocking on exam year: candidate pairs were formed either when the school the child reported attending in a given year matched the KCPE school for that year, or, more broadly, when any school ever reported for the child matched the KCPE school and the exam year fell within the child's plausible KCPE year range. This step limited the pool of potential matches for each child to those who sat for the KCPE exam at the same school.

Step 5: Matching names within schools. Within the restricted set of schools, we compared the cleaned and tokenized names from KLPS and KCPE. We measured overlap between the sets of tokens, while allowing for small differences in spelling using fuzzy string matching: tokens were considered a match at a Jaro–Winkler similarity of 0.86 or higher. Candidate matches were then scored based on two factors: (1) the share of KCPE tokens that appeared in the survey names, and (2) whether there was an exact or close school match. The scoring was designed so that a perfect coverage of the KCPE name tokens combined with a school agreement would yield the maximum possible score.

Step 6: Identifying perfect matches. A match was considered "perfect" if the survey names captured all tokens in the KCPE record and the school matched. These are high-confidence matches. If a respondent had exactly one such perfect match, it was accepted automatically. When multiple perfect matches occurred for the same respondent, additional filters were needed to resolve them.

Step 7: Applying additional filters for multiple matches. Two further checks were introduced. First, we compared gender: matches where the KCPE record listed a sex that directly contradicted the survey information were dropped. Second, we checked whether the exam year was consistent with the respondent's reported grade progression in KLPS. Specifically, we estimated the expected KCPE year based on the grade the child reported being in at baseline and compared this with the official exam year. Matches that were implausibly early were removed, and if more than one plausible record remained, we selected the earliest KCPE year consistent with the child's schooling trajectory.

Step 8: Resolving collisions across respondents In some cases, multiple KLPS respondents matched to the same KCPE record. These "collisions" required additional adjudication. We compared how closely each respondent's expected exam year aligned with the

actual KCPE year. If only one respondent fell within a small acceptable margin (within zero to two years), that respondent was selected.

C Additional Details on the Validity of the Design

Stability of the cutoffs over time. Because both the KCPE scale and the examination itself changed across years, Appendix Figure A3 repeats the structural break search separately for exam years 1999–2000, 2001–2002, and 2003–2006. These searches are much noisier than the pooled search, because each gender-by-period cell contains only about 280 to 620 observations within the search window, compared with about 1,100 to 1,400 in the pooled search, and the resulting R^2 profiles are correspondingly flat. Within the range of 225 to 275 points, the female search peaks at 234 in 1999–2000 and 2001–2002 and at 239 in 2003–2006. The male search peaks at 251 in 1999–2000, matching the pooled search, and at 243 and 227 in the later periods, where the profiles are flatter. Where the maximizing cutoff falls outside this range, most visibly for men in 2003–2006, the profile is nearly flat, so the subsample is uninformative about the location of the threshold rather than pointing to a different one.

Match rates. Appendix Table A4 estimates the discontinuity in the probability that a KCPE archive record is matched to a KLPS respondent. The level of this match rate is mechanically low, because the archive includes candidates from other cohorts and repeat attempts that could not be linked to the KLPS sample. Even if every KLPS respondent who reached Standard 8 were linked, at most 30 percent of records could be matched, and just below the cutoff the procedure attains about three-quarters of that ceiling. The object of interest is therefore the discontinuity rather than the level. Match rates are smooth at the cutoff in the pooled sample and among male examinees, but fall by 5 to 6 percentage points among female examinees. Retaking was common in this cohort. Among the respondents Ozier (2018) matched to official records, about 10 percent had scores from two exam years, typically consecutive. Near-miss candidates who retook the exam could cross the threshold on their second attempt. Because the matching procedure links each respondent to the earliest exam record consistent with her schooling history, these later attempts appear in the archive as unmatched records concentrated just above the cutoff. Consistent with this interpretation, apparent repeat records are almost never matched (2.5 percent, compared with 19 percent overall) and lie overwhelmingly above the cutoff, and the female discontinuity shrinks and loses significance when the score points nearest the threshold are excluded (donut estimates of -0.024 to -0.039, with robust p-values of 0.18 to 0.40). Ozier (2018) found a related deficit in the density of matched women just above the cutoff (McCrary $p = 0.017$) and noted it as a reason for caution in interpreting the results for women. In the expanded sample used here, the density of matched respondents is smooth at the cutoff for both women ($p = 0.92$) and men ($p = 0.74$), and covariates are balanced among matched women (Appendix Table A2).